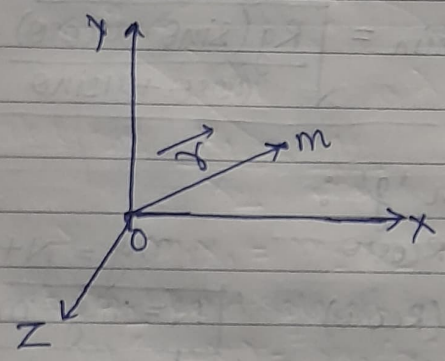
Centre of mass

→ It is point representation of a large body.

→ It is a point at which entire mass of large body can be assumed to be concentrated and if Fext is applied on it, it will have same translational motion as that of the large body.

→ It can be real or imaginary point.

Mass moment :

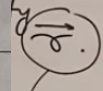


$$\text{Mass moment} = m\vec{r}$$

Unit = Kg-m Dimension = [ML]

Vector quantity

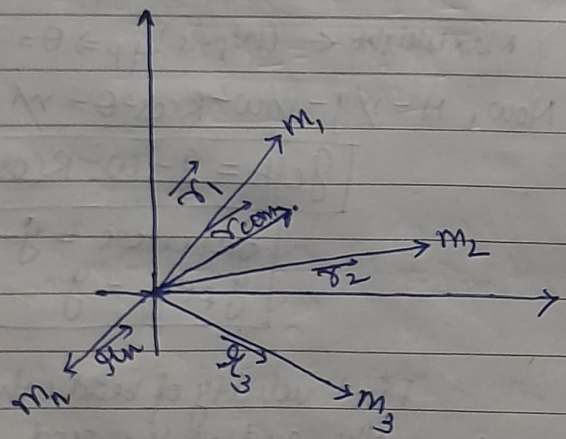
Direction along $\vec{r}$



↳ object: Head

↳ obs. : Tail

COM of a system of discrete particles :



$$\vec{r}_{com} = \frac{\text{Summation of mass moment}}{\text{Summation of mass}}$$

$$\vec{r}_{com} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \dots + m_n\vec{r}_n}{m_1 + m_2 + \dots + m_n}$$

$$\vec{r}_{com} = \frac{\sum_{i=1}^{n} m_i\vec{r}_i}{\sum_{i=1}^{n} m_i}$$

Q 1gm (2,3,4)m
2gm (1,-1,2)m
2gm (2,0,-2)m

Find $\vec{r}_{com}$ of this system

$$\vec{r}_{com} = \frac{1(2\hat{i}+3\hat{j}+4\hat{k}) + 2(\hat{i}-\hat{j}+2\hat{k}) + 2(2\hat{i}+0\hat{j}-2\hat{k})}{1+2+2}$$

$$\vec{r}_{com} = \frac{2\hat{i}+3\hat{j}+4\hat{k}+2\hat{i}-2\hat{j}+4\hat{k}+4\hat{i}+0\hat{j}-4\hat{k}}{5}$$

$$\vec{r}_{com} = \frac{8\hat{i}+\hat{j}+4\hat{k}}{5}\, m = \left(\frac{8}{5}, \frac{1}{5}, \frac{4}{5}\right) m$$

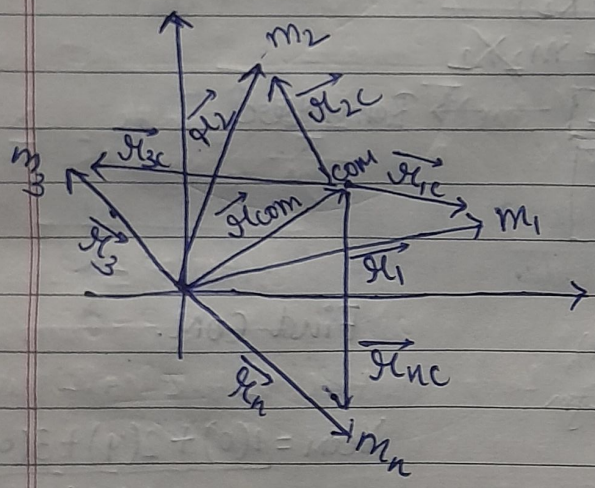
Note:

① $X_{com} = \dfrac{m_1x_1 + m_2x_2 + m_3x_3 + \cdots + m_nx_n}{m_1+m_2+m_3+\cdots+m_n}$

② $Y_{com} = \dfrac{m_1y_1 + m_2y_2 + m_3y_3 + \cdots + m_ny_n}{m_1+m_2+m_3+\cdots+m_n}$

③ $Z_{com} = \dfrac{m_1z_1 + m_2z_2 + m_3z_3 + \cdots + m_nz_n}{m_1+m_2+m_3+\cdots+m_n}$

* **Ex** Prove that summation of mass moments wrt COM is zero.



Sum of mass moments wrt COM

$= m_1\vec{r}_{1C} + m_2\vec{r}_{2C} + \cdots + m_n(\vec{r}_{nC})$

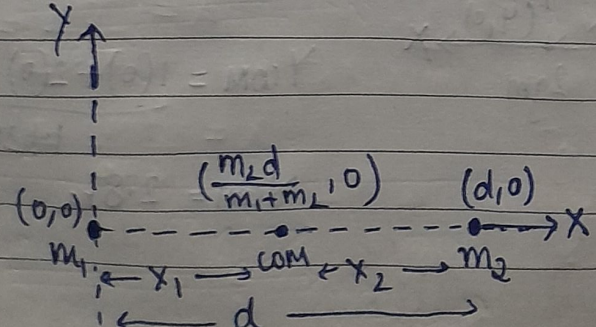
$= m_1(\vec{r}_1 - \vec{r}_C) + m_2(\vec{r}_2 - \vec{r}_C) + \cdots + m_n(\vec{r}_n - \vec{r}_C)$

$= (m_1\vec{r}_1 + m_2\vec{r}_2 + \cdots + m_n\vec{r}_n) - \vec{r}_C(m_1+m_2+\cdots+m_n)$

$= (m_1\vec{r}_1 + \cdots + m_n\vec{r}_n) - \dfrac{(m_1\vec{r}_1 + \cdots + m_n\vec{r}_n)(m_1+\cdots+m_n)}{(m_1+m_2+\cdots+m_n)}$

$= 0$ (Null vector)

COM of two point masses:



$$X_{COM} = \frac{m_1(0)+m_2(d)}{m_1+m_2}$$

$$\boxed{X_{COM} = \frac{m_2 d}{m_1+m_2}}$$

$$\boxed{Y_{COM} = \frac{m_1(0)+m_2(0)}{m_1+m_2} = 0}$$

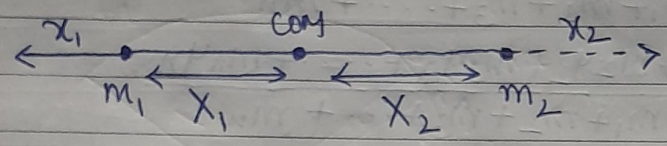
Let the distance of COM from m_1 be X_1 and from m_2 be X_2

$\Rightarrow$ $\boxed{X_1 = \frac{m_2 d}{m_1+m_2}}$ $\quad$ $\boxed{X_2 = d - X_1 = d - \frac{m_2 d}{m_1+m_2} = \frac{m_1 d}{m_1+m_2}}$

$\therefore$ $\boxed{\frac{X_1}{X_2} = \frac{m_2}{m_1}}$ $\Rightarrow$ $\boxed{X \propto \frac{1}{m}}$ $\Rightarrow$ COM will lie nearer to heavy object and farther from lighter object.

$\rightarrow$ COM lies on line segment joining m_1 and m_2.

Q



m_1 is displaced by distance x_1 as given in diagram. Find displacement of m_2, such that COM remains at same position.

We know, $\frac{X_1}{X_2} = \frac{m_2}{m_1}$

$\Rightarrow$ $m_1 X_1 = m_2 X_2$ —— (1)

Let m_2 be displaced by x_2 to the right.

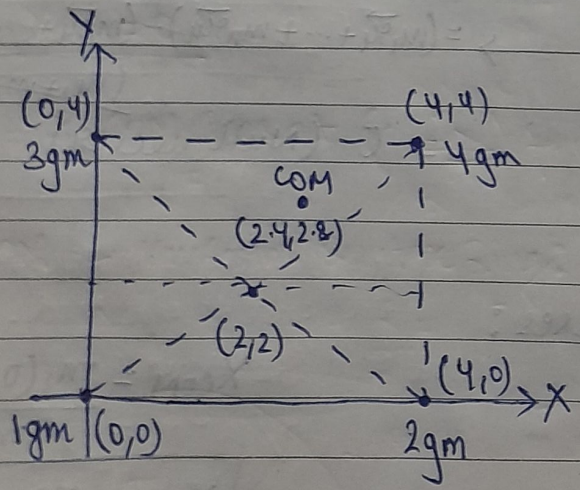
$m_1(x_1+X_1) = m_2(x_2+X_2)$

$m_1 x_1 + m_1 X_1 = m_2 x_2 + m_2 X_2$

$\boxed{m_1 x_1 = m_2 x_2}$ $\longrightarrow$ Remember

$\therefore$ $\boxed{x_2 = \frac{m_1 x_1}{m_2}}$

Q



Find COM.

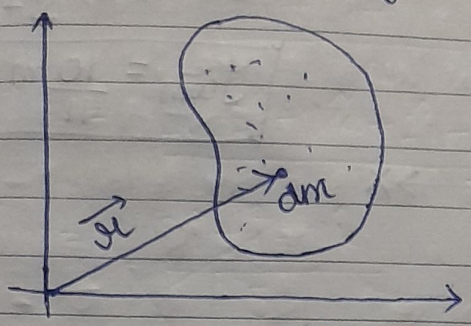
$X_{COM} = \frac{1(0)+2(4)+3(0)+4(4)}{1+2+3+4}$

$X_{COM} = 2.4m$

$Y_{COM} = \frac{1(0)+2(0)+3(4)+4(4)}{1+2+3+4}$

$Y_{COM} = 2.8m$

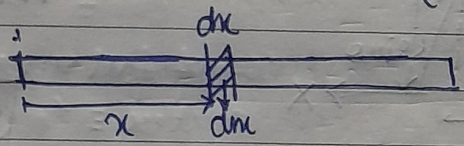
COM of Continuous Body:



$$\vec{r}_{COM} = \frac{\int \vec{r}\, dm}{\int dm}$$

$$x_{COM} = \frac{\int x\, dm}{\int dm} \qquad y_{COM} = \frac{\int y\, dm}{\int dm}$$

① COM of uniform Rod:



$$x_{COM} = \frac{\int x\, dm}{\int dm} = \frac{\lambda \int_0^l x\, dx}{m}$$

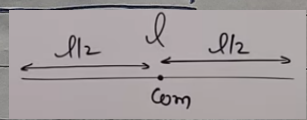
$$= \frac{\lambda \left(\frac{x^2}{2}\right)_0^l}{m} = \frac{\left(\frac{m}{l}\right)\left(\frac{l^2}{2}\right)}{m}$$

$\lambda = \frac{m}{l}$ for complete rod.

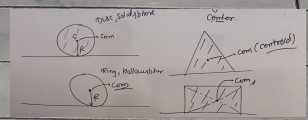
$\lambda = \frac{dm}{dx} \Rightarrow \boxed{dm = \lambda\, dx}$

$$\boxed{x_{COM} = \frac{l}{2}} \qquad \boxed{y_{COM} = 0}$$

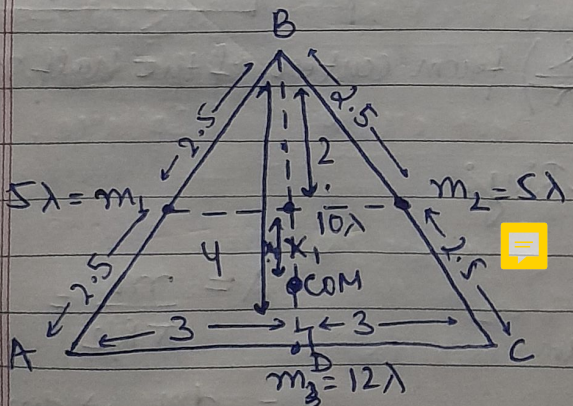
∴ COM of uniform rod lies at $\left(\frac{l}{2}, 0\right)$



→ In a symmetric system, if mass distribution is uniform, then COM lies at geometrical center.



Q



Given 3 uniform wire

AB = 5m, λ

BC = 5m, λ

AC = 6m, 2λ

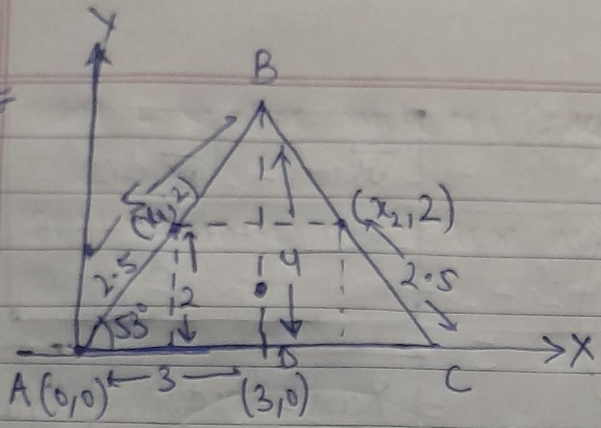
Find position of their COM

$$x_1 = \frac{12\lambda}{10\lambda + 12\lambda} \times 2 = \frac{12}{11} m$$

COM lies on BD → Distance from B $= 2 + x_1 = \frac{34}{11} m$

Distance from D $= 2 - x_1 = \frac{10}{11} m$

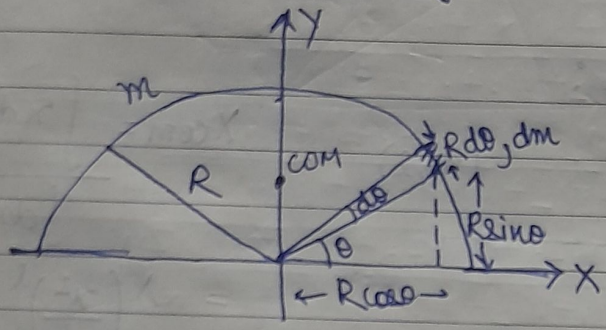
Alternate



$$y_{com} = \frac{5\lambda(2) + 5\lambda(2) + 12\lambda(0)}{5+5+12}$$

$$\boxed{y_{com} = \frac{10}{11}\,m} \quad \text{from D}$$

② COM of Uniform Half Ring:



$$\lambda = \frac{m}{\pi R}$$

$$\lambda = \frac{dm}{Rd\theta} \Rightarrow dm = \lambda R d\theta = \frac{m}{\pi R} R\, d\theta$$

$$\boxed{dm = \frac{m}{\pi} d\theta}$$

$$x_{com} = \frac{\int x\,dm}{\int dm} = \frac{\int R\cos\theta \left(\frac{m}{\pi} d\theta\right)}{m}$$

$$= \frac{\frac{mR}{\pi}\int_0^{\pi} \cos\theta\, d\theta}{m} = \frac{R}{\pi}\left(\sin\theta\right)_0^{\pi}$$

$$\boxed{x_{com} = 0}$$

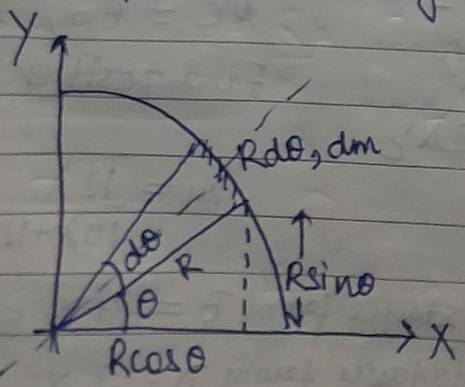
$$y_{com} = \frac{\int y\,dm}{\int dm} = \frac{\int R\sin\theta \left(\frac{m}{\pi} d\theta\right)}{m}$$

$$= \frac{\frac{mR}{\pi}\int_0^{\pi} \sin\theta\, d\theta}{m} = \frac{R}{\pi}\left(-\cos\theta\right)_0^{\pi}$$

$$= \boxed{y_{com} = \frac{2R}{\pi}}$$

∴ COM lies at $\left(0, \frac{2R}{\pi}\right)$ from center of the half ring.

③ COM of Uniform Quater Ring:



$$\lambda = \frac{m}{\pi R/2} = \frac{2m}{\pi R}$$

$$\lambda = \frac{dm}{Rd\theta} \Rightarrow dm = \lambda R d\theta = \frac{2m}{\pi R} R\, d\theta$$

$$\boxed{dm = \frac{2m}{\pi} d\theta}$$

$$X_{COM} = \frac{\int x\,dm}{\int dm} = \frac{\int R\cos\theta \left(\frac{2m}{\pi}\,d\theta\right)}{m}$$

$$= \frac{2mR}{\pi} \int \cos\theta\, d\theta \Big/ m = \frac{2R}{\pi} \int_0^{\pi/2} \cos\theta\, d\theta$$

$$= \boxed{X_{COM} = \frac{2R}{\pi}}$$

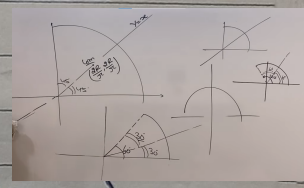
$$Y_{COM} = \frac{\int y\,dm}{\int dm} = \frac{\int R\sin\theta\, dm}{m}$$

$$= \int \frac{R\sin\theta \left(\frac{2m}{\pi}\,d\theta\right)}{m}$$

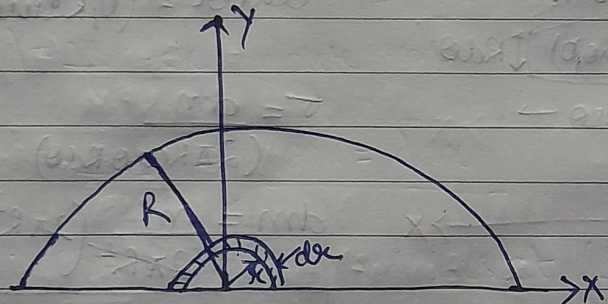
$$= \frac{2R}{\pi} \int_0^{\pi/2} \sin\theta\, d\theta$$

$$\boxed{Y_{COM} = \frac{2R}{\pi}}$$

$\therefore$ COM lies at $\left(\frac{2R}{\pi}, \frac{2R}{\pi}\right)$ from centre of ring.
Also COM lies on line $y = x$.



(4) COM of Uniform Half Disc: (σ = mass/Area)



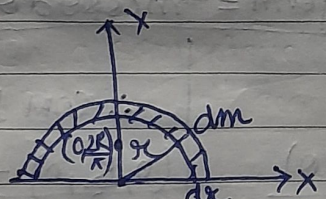
$$\sigma = \frac{2m}{\pi R^2}$$

$$\sigma = \frac{dm}{\pi x\,dx}$$

$$dm = \sigma \pi x\,dx$$

$$= \frac{2m}{\pi R^2} \pi x\,dx$$

$$\boxed{dm = \frac{2mx\,dx}{R^2}}$$



COM of elemental ring $(x, y) \equiv \left(0, \frac{2x}{\pi}\right)$

$$X_{COM} = \frac{\int x\,dm}{\int dm} = \frac{\int 0\,dm}{m}$$

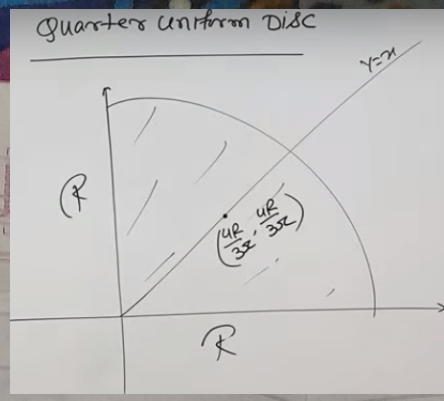
$$\boxed{X_{COM} = 0}$$

$$Y_{COM} = \frac{\int y\,dm}{\int dm} = \frac{\int \frac{2x}{\pi}\left(\frac{2mx\,dx}{R^2}\right)}{m}$$

$$= \frac{4}{\pi R^2} \int_0^R x^2\,dx = \frac{4}{\pi R^2}\left(\frac{R^3}{3}\right)$$

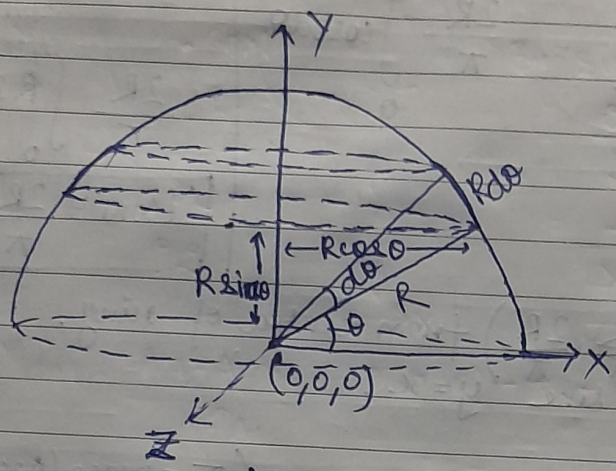
$$\boxed{Y_{COM} = \frac{4R}{3\pi}}$$

$\therefore$ COM lies at $\left(0, \frac{4R}{3\pi}\right)$ from centre of disc.

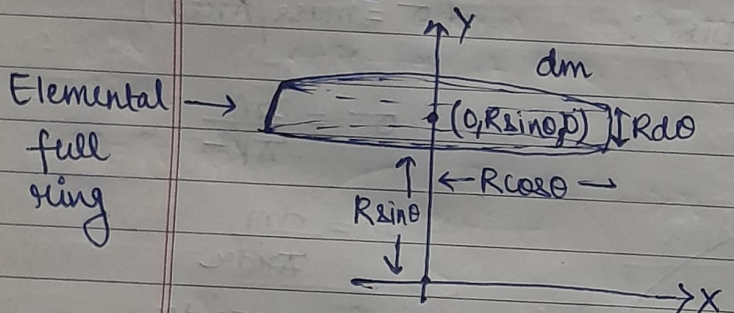


COM of Quater uniform disc = $\left(\frac{4R}{3\pi}, \frac{4R}{3\pi}\right)$

⑤ COM of uniform hollow hemisphere:



$\sigma = \frac{m}{2\pi R^2}$.



COM of full ring lies at its center $\equiv (0, R\sin\theta, 0)$

$\sigma = \frac{dm}{(2\pi R\cos\theta \cdot Rd\theta)}$

$dm = \frac{m}{2\pi R^2}(2\pi R^2 \cos\theta\, d\theta)$

$\boxed{dm = m\cos\theta\, d\theta}$

$X_{COM} = \frac{\int x\,dm}{\int dm}$

$= \frac{\int 0\,dm}{m} = 0$

$\boxed{X_{COM} = 0}$

$Y_{COM} = \frac{\int y\,dm}{\int dm}$

$= \int \frac{R\sin\theta (m\cos\theta\, d\theta)}{m}$

$= \cancel{2}R \int_0^{\pi/2} \sin\theta\cos\theta\, d\theta$

$= \frac{R}{2} \int_0^{\pi/2} 2\sin\theta\cos\theta\, d\theta$

$= \frac{R}{2} \int_0^{\pi/2} \sin 2\theta\, d\theta$

$= \frac{R}{2}\left(\frac{-\cos 2\theta}{2}\right)_0^{\pi/2}$

$\boxed{Y_{COM} = R/2}$

$Z_{COM} = \frac{\int z\,dm}{\int dm}$

$= \frac{\int 0\,dm}{m} = 0$

$\boxed{Z_{COM} = 0}$

∴ COM lies at $(0, \frac{R}{2}, 0)$ from centre of hollow hemisphere.

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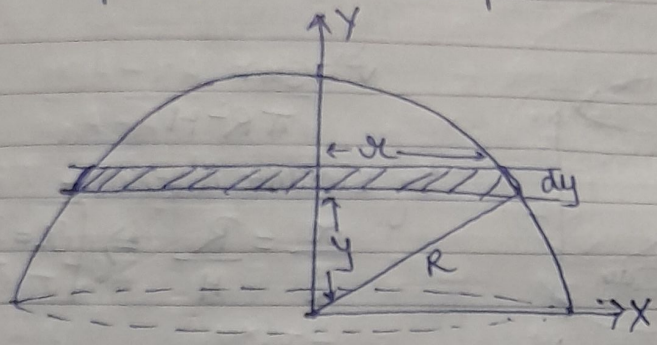
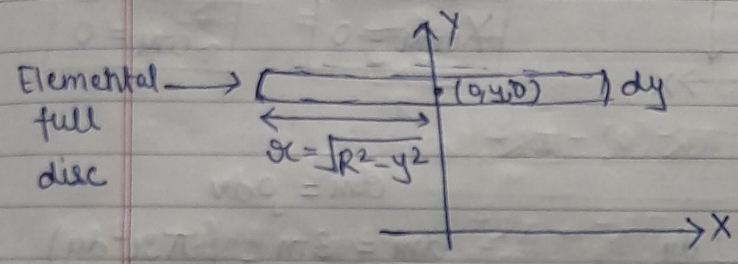
⑥ COM of Uniform solid hemisphere : (ρ = mass/Volume)



$$\rho = \frac{3m}{2\pi R^3}$$



COM of full disc lies at its center $\equiv (0, y, 0)$

$$\rho = \frac{dm}{\pi(R^2 - y^2)dy}$$

$$dm = \frac{3m}{2\pi R^3} \times \pi (R^2 - y^2)dy$$

$$\boxed{dm = \frac{3m}{2R^3}(R^2 - y^2)dy}$$

$$\boxed{X_{com} = 0}$$

$$\boxed{Z_{com} = 0}$$

$$Y_{com} = \frac{\int y\,dm}{\int dm} = \frac{\int y\left(\frac{3m}{2R^3}(R^2 - y^2)dy\right)}{m} = \frac{3}{2R^3}\int y(R^2 - y^2)dy$$

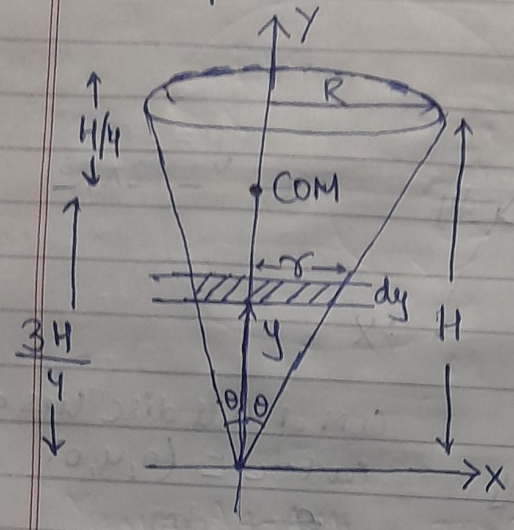
$$= \frac{3}{2R^3}\int_0^R (R^2 y - y^3)dy = \frac{3}{2R^3}\left(\frac{R^2 y^2}{2} - \frac{y^4}{4}\right)_0^R$$

$$= \frac{3}{2R^3}\left(\frac{R^4}{2} - \frac{R^4}{4}\right) = \frac{3}{2R^3}\left(\frac{R^4}{4}\right) = \frac{3R}{8} \quad \therefore \boxed{Y_{com} = \frac{3R}{8}}$$

$\therefore$ COM lies at $\left(0, \frac{3R}{8}, 0\right)$ from centre of solid hemisphere.

Alternate: we can use an elemental hollow hemisphere and integrate.

(7) COM of Solid Cone:



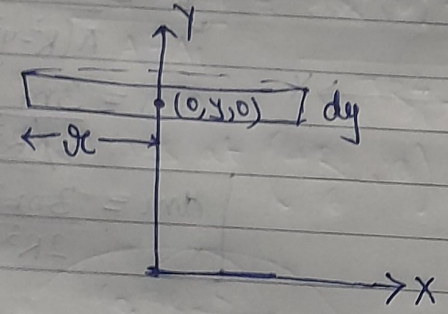
$$\rho = \frac{3m}{\pi R^2 H}$$

By similar Δ's,

$$\frac{x}{y} = \frac{R}{H} \Rightarrow \boxed{x = \frac{Ry}{H}}$$

$$\boxed{X_{COM} = 0} \qquad \boxed{Z_{COM} = 0}$$

Elemental disc → full disc



COM = (0, y, 0)

$$dm = \rho dv$$

$$dm = \frac{3m}{\pi R^2 H}(\pi x^2 dy)$$

$$= \frac{3m}{\pi R^2 H}\left(\pi \frac{R^2 y^2}{H^2} dy\right)$$

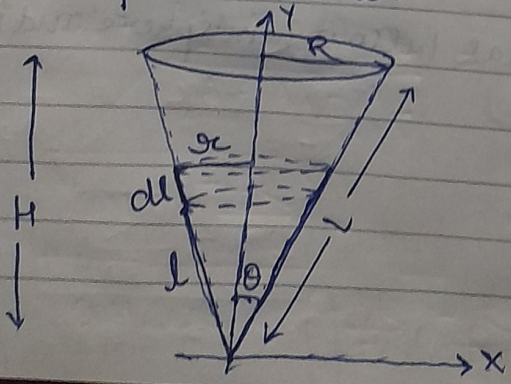
$$\boxed{dm = \frac{3m}{H^3} y^2 dy}$$

$$Y_{COM} = \frac{\int dm(y)}{\int dm} = \frac{\int y\left(\frac{3m}{H^3} y^2\right) dy}{m} = \frac{3}{H^3}\int_0^H y^3 dy$$

$$= \frac{3}{H^3}\left(\frac{H^4}{4}\right) = \frac{3H}{4} \qquad \therefore \boxed{Y_{COM} = \frac{3H}{4}}$$

∴ COM lies at $(0, \frac{3H}{4}, 0)$ from top of the cone and at $(0, \frac{H}{4}, 0)$ from base of the cone.

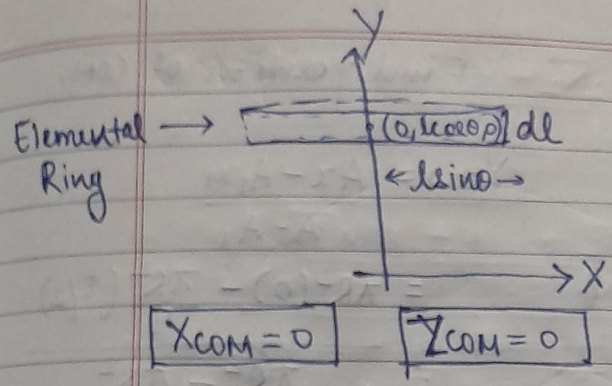
(8) COM of Hollow Cone:



$$\sigma = \frac{m}{\pi R L}$$

$$R = L\sin\theta, \quad H = L\cos\theta$$

$$x = l\sin\theta$$



$X_{COM} = 0$ $\quad$ $Y_{COM} = 0$

$dm = \sigma\, dA$

$dm = \frac{m}{\pi RL} \times 2\pi l \sin\theta\, dl$

$$dm = \frac{2m}{RL} l \sin\theta\, dl$$

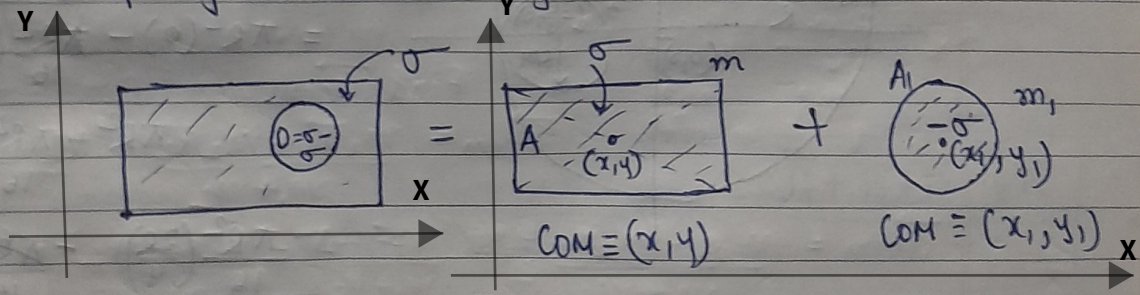
$$Y_{COM} = \frac{\int y\, dm}{\int dm} = \frac{\int (l\cos\theta)\left(\frac{2m}{RL} l\sin\theta\, dl\right)}{m} = \frac{2}{RL}\sin\theta\cos\theta \int_0^L l^2\, dl$$

$$= \frac{2}{RL}\sin\theta\cos\theta \int_0^L l^2\, dl = \frac{2}{RL}\left(\frac{R}{L}\right)(\cos\theta)\left(\frac{L^3}{3}\right) = \frac{2}{3} L\cos\theta = \frac{2}{3}H$$

$$Y_{COM} = \frac{2}{3} L\cos\theta = \frac{2}{3} H$$

∴ COM lies at $(0, \frac{2}{3}H, 0)$ from top and $(0, \frac{H}{3}, 0)$ from base of the cone

COM of objects with Cavity:



$$X_{COM} = \frac{mx + m_1 x_1}{m + m_1}$$

$$= \frac{\sigma A x + (-\sigma) A_1 x_1}{\sigma A + (-\sigma)(A_1)}$$

$$X_{COM} = \frac{Ax - A_1 x_1}{A - A_1}$$

Similarly, $Y_{COM} = \frac{Ay - A_1 y_1}{A - A_1}$

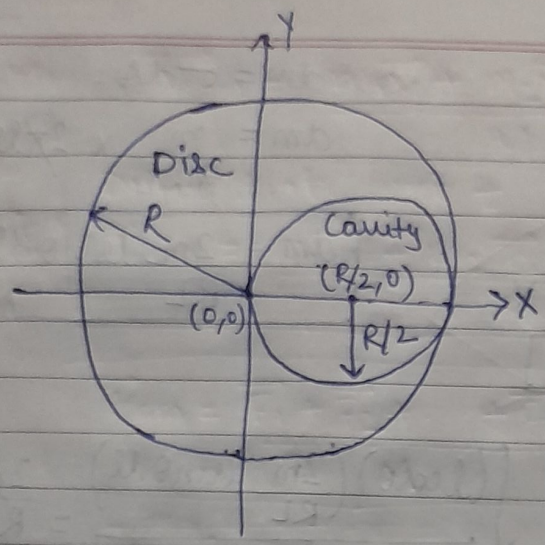
A = Area without cavity

(x, y) ≡ COM of system without cavity

A_1 = Area of cavity

(x_1, y_1) = COM of cavity

Q



Find position of COM.

$$X_{COM} = \frac{Ax - A_1x_1}{A - A_1}$$

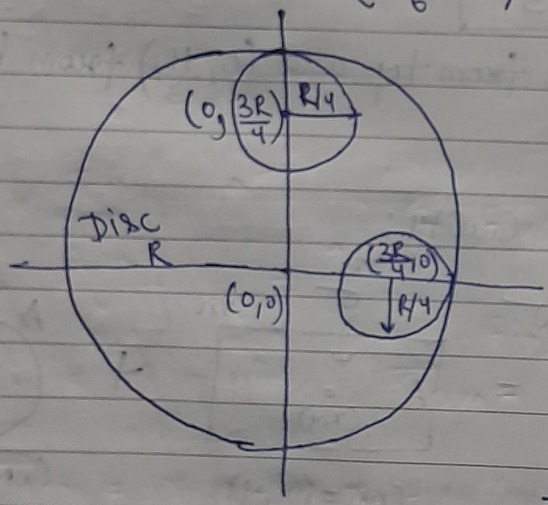
$$= \frac{\pi R^2(0) - \frac{\pi R^2}{4}(R/2)}{\pi R^2 - \frac{\pi R^2}{4}}$$

$$= \frac{-\frac{\pi R^3}{8}}{\frac{3\pi R^2}{4}} = \frac{-R}{6}$$

$$Y_{COM} = \frac{Ay - A_1y_1}{A - A_1} = \frac{\pi R^2(0) - \frac{\pi R^2}{4}(0)}{\pi R^2 - \frac{\pi R^2}{4}} = 0$$

$\therefore$ COM lies at $\left(-\frac{R}{6}, 0\right)$

Q



Find position of COM

$$X_{COM} = \frac{Ax - A_1x_1 - A_2x_2}{A - A_1 - A_2}$$

$$= \frac{\pi R^2(0) - \frac{\pi R^2}{16}\left(\frac{3R}{4}\right) - \frac{\pi R^2}{16}(0)}{\pi R^2 - \frac{\pi R^2}{16} - \frac{\pi R^2}{16}}$$

$$= \frac{-\frac{\pi R^2}{16}\left(\frac{3R}{4}\right)}{\frac{7\pi R^2}{8}} = \frac{-\frac{\pi R^2}{2\cancel{16}}\left(\frac{3R}{4}\right)}{\frac{7\pi R^2}{8}} = \frac{-3R}{56}$$

$$Y_{COM} = \frac{Ay - A_1y_1 - A_2y_2}{A - A_1 - A_2} = \frac{\pi R^2(0) - \frac{\pi R^2}{16}(0) - \frac{\pi R^2}{16}\left(\frac{3R}{4}\right)}{\pi R^2 - \frac{\pi R^2}{16} - \frac{\pi R^2}{16}}$$

$$\boxed{Y_{COM} = \frac{-3R}{56}}$$

$\therefore$ COM lies at $\left(\frac{-3R}{56}, \frac{-3R}{56}\right)$ $\{y = x$ line$\}$

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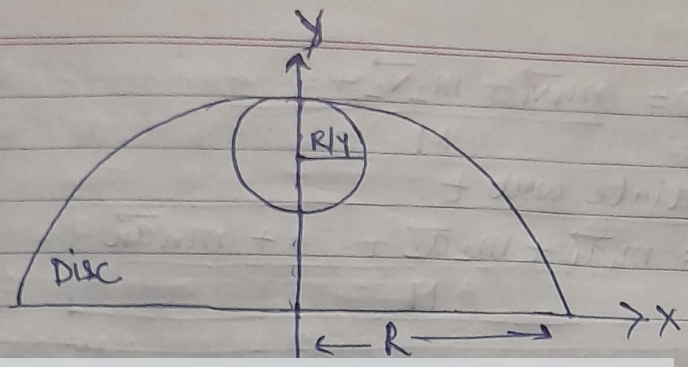
Q



Find position of COM.

COM of Half disc = $\left(0, \frac{4R}{3\pi}\right)$

$$X_{com} = \frac{\frac{\pi R^2}{2}(0) - \frac{\pi R^2}{16}(0)}{\frac{\pi R^2}{2} - \frac{\pi R^2}{16}} = 0$$

$$Y_{com} = \frac{\frac{\pi R^2}{2}\cdot\frac{4R}{3\pi} - \frac{\pi R^2}{16}\times\frac{3R}{4}}{\frac{\pi R^2}{2} - \frac{\pi R^2}{16}}$$

$\frac{56x - 9\pi x}{180\pi}$

Motion of COM:

$$\vec{R}_{COM} = \frac{m_1\vec{R_1} + m_2\vec{R_2} + \dots + m_n\vec{R_n}}{m_1 + m_2 + m_3 \dots + m_n}$$

Displacement of COM

$$d\vec{R}_{com} = \frac{m_1 d\vec{R_1} + m_2 d\vec{R_2} + \dots}{m_1 + m_2 + \dots} \Rightarrow \Delta R_{com} = \frac{m_1\Delta R_1 + m_2\Delta R_2 + \dots}{m_1 + m_2}$$

Differentiate wrt time.

$$\Delta R_{com} = \frac{m_1\Delta R_1 + m_2\Delta R_2 + \dots}{m_1 + m_2 + \dots}$$

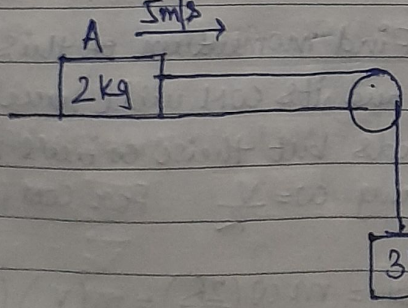
$(P = mV)$

Velocity of COM

$$\vec{V}_{com} = \frac{m_1\vec{V_1} + m_2\vec{V_2} + \dots + m_n\vec{V_n}}{m_1 + m_2 + \dots + m_n} = \frac{\vec{P_1} + \vec{P_2} + \vec{P_3} \dots + \vec{P_n}}{M}$$

$$\vec{V}_{com} = \frac{\vec{P}_{net}}{M} \Rightarrow \boxed{\vec{P}_{net} = M\vec{V}_{com}}$$ (Total momentum of system)

Q



Find $\vec{V}_{com}$ of system (A+B)

$$\vec{V}_{com} = \frac{2(5\hat{i}) + 3(-5\hat{j})}{5}$$

$$= (2\hat{i} - 3\hat{j})\ m/s$$

$$\vec{P}_{net} = 5(2\hat{i} - 3\hat{j}) = 10\hat{i} - 15\hat{j}$$ Kgm/s

Ex

find displacement of COM

$$\Delta\vec{R}_{com} = \frac{2gm\,10cm\,\hat{i} + 5gm(-2cm\,\hat{j})}{7gm}$$

$$\Delta\vec{R}_{com} = \left(\frac{20}{7}\hat{i} - \frac{10}{7}\hat{j}\right)cm$$

$$\vec{V}_{com} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2 + \cdots}{M}$$

Differentiate wrt t.

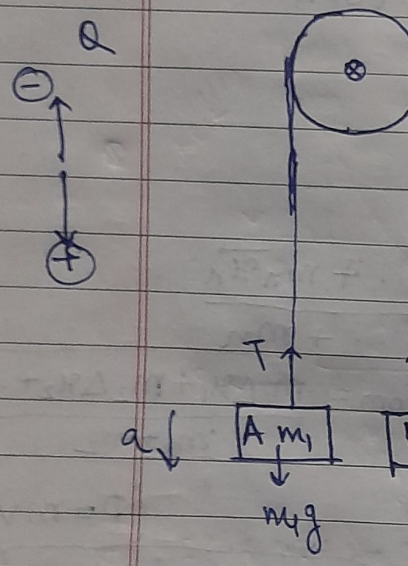
acceleration of COM

$$\boxed{\vec{a}_{com} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2 + \cdots + m_n\vec{a}_n}{M}}$$

$$\boxed{\vec{a}_{com} = \frac{\vec{F}_1 + \vec{F}_2 + \cdots + \vec{F}_n}{M} = \frac{\vec{F}_{net}}{M} = \frac{\vec{F}_{ext}}{M}}$$

$$\therefore \boxed{a_{com} = \frac{\vec{F}_{ext}}{M}}$$

Q



$m_1 > m_2$, System = A+B

① Find acc^n of Block A and B

$$\boxed{a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)g}$$

② Find Tension: $T = \left(\frac{2m_1m_2}{m_1+m_2}\right)g$

③ Find $\vec{a}_{com}$.

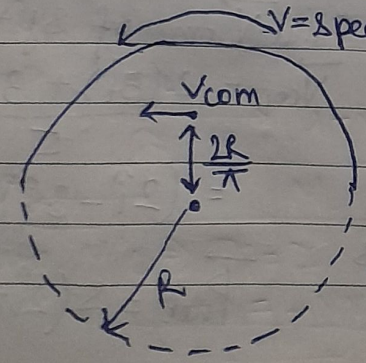
$$\vec{a}_{com} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1+m_2} = \frac{m_1a + m_2(-a)}{m_1+m_2}$$

$$\vec{a}_{com} = \frac{m_1a - m_2a}{m_1+m_2} = a\left(\frac{m_1-m_2}{m_1+m_2}\right) = \left(\frac{m_1-m_2}{m_1+m_0}\right)g\left(\frac{m_1-m_2}{m_1+m_2}\right)$$

$$\therefore \boxed{\vec{a}_{com} = \left(\frac{m_1-m_2}{m_1+m_2}\right)^2 g}$$

$$a_{com} = \frac{f_{ext}}{M} = \frac{(m_1+m_2)g - 2T}{(m_1+m_2)}$$

Q



Find momentum of this half ring

ans. Ring and its COM will have different speeds but their ω will be same.

For ring $\omega = \frac{V}{R}$ For COM, $\omega = \frac{V_{com}}{2R/\pi}$

$$\vec{P} = MV_{com} = m\omega\left(\frac{2R}{\pi}\right) = m\left(\frac{V}{R}\right)\left(\frac{2R}{\pi}\right) = \frac{2mV}{\pi}$$

Linear Momentum Conservation Theorem: If $\vec{F}_{ext}$ on a system is zero, then linear momentum of system is constant or conserved.

If $\vec{F}_{ext}=0 \Rightarrow \vec{a}_{com}=0 \Rightarrow \vec{V}_{com}=$ const

$\vec{P}_{system} = M\vec{V}_{com} =$ const

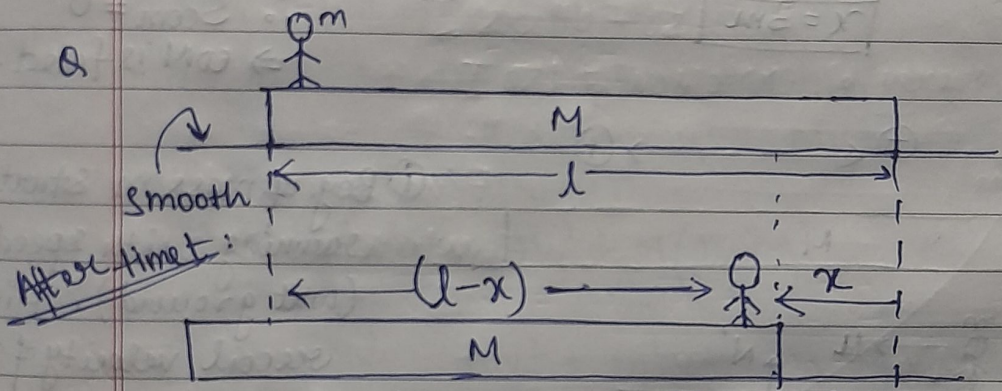
$\therefore \boxed{\vec{P}_i = \vec{P}_f}$

If $(\vec{F}_{ext})_x = 0 \Rightarrow (\vec{P}_i)_x = (\vec{P}_f)_x$

If $(F_{ext})_y = 0 \Rightarrow (P_i)_y = (P_f)_y$

* If COM is fixed $\Rightarrow \vec{V}_{com}=0 \Rightarrow \vec{a}_{com}=0$

Q



Find displacement of planck in the time, in which boy reaches the other end of planck

COM of the system will remain fixed.

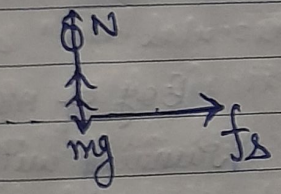
$\Rightarrow m_1x_1 = m_2x_2$

$Mx = m(l-x)$

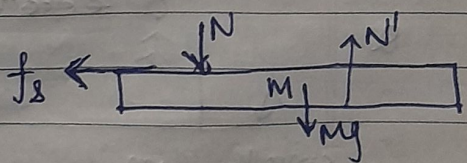
$\boxed{x = \frac{ml}{M+m}}$

Concept

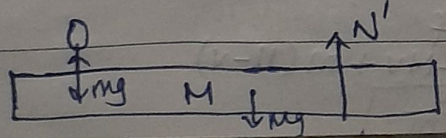
For Boy, Momentum will ↑ in +ve x due to f_s



For Planck, P will ↑ in -ve x



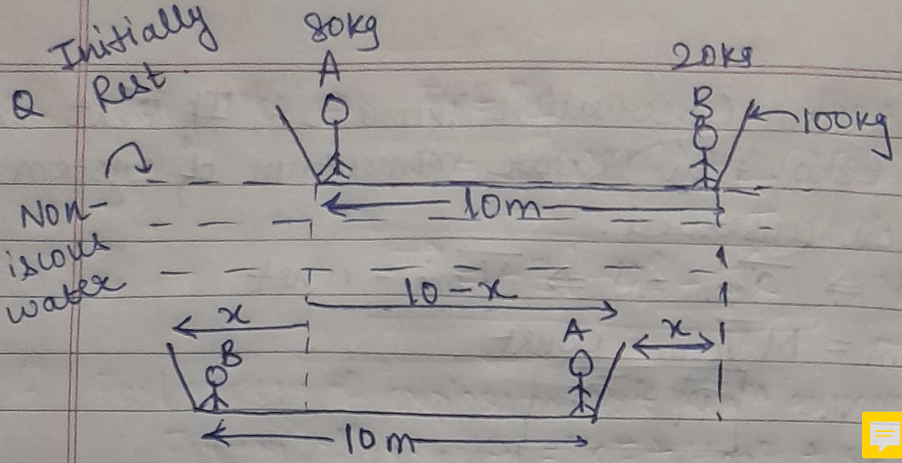
But for system (M+m), $\vec{P}$ is conserved in x direction as f_s is internal force and hence no $\vec{F}_{ext}$ exists.



$a_{com} = 0$

$a_{com} = 0$

Q Initially Rest



COM is fixed ⇒

$100x + 20(x+10) = 80(10-x)$

$10x + 2x + 20 = 80 - 8x$

$20x = 60$

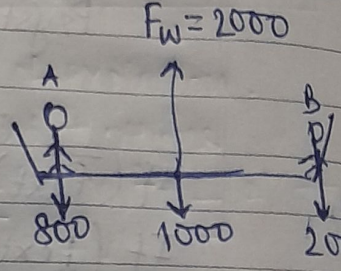
$\boxed{x = 3m}$

$S_{Boat} = 3m \leftarrow$

$S_A = 7m \rightarrow$

$S_B = 13m \leftarrow$

Find displacement of boat, if boys interchange their positions on the boat



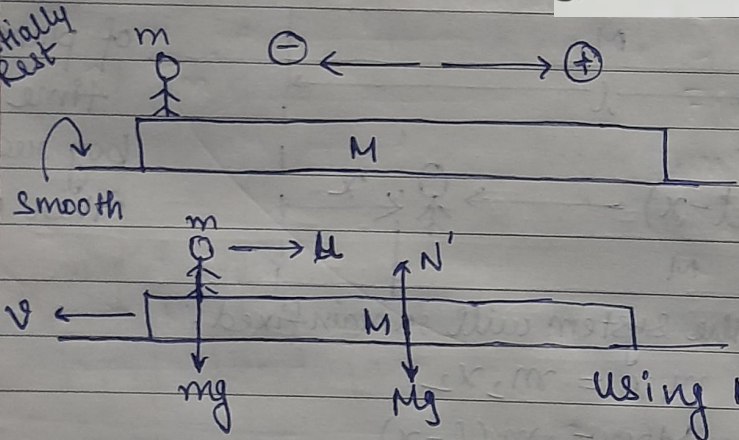
$U_{com} = 0, \Rightarrow F_{ext} = 0$

$\Rightarrow \vec{a}_{com} = 0$

$\therefore \vec{S}_{com} = 0$

$\Rightarrow$ COM is fixed.

Q Initially Rest



① Boy of mass m starts running with speed u (wrt ground). Find recoil velocity of Plank

$F_{ext} = 0$

Using M/C ⇒ $P_i = P_f$

$0 + 0 = -Mv + mu$

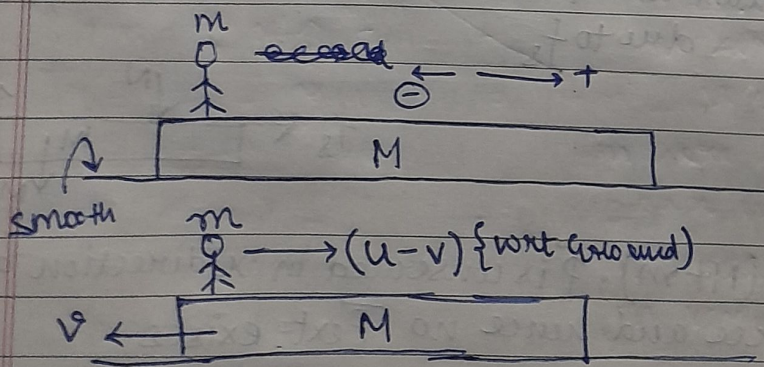
$\boxed{v = \frac{mu}{M}}$

② Find W.d by boy.

W.d. by boy = $K_f - K_i$

$= \frac{1}{2}mu^2 + \frac{1}{2}M\left(\frac{mu}{M}\right)^2 - 0$

Q



Boy of mass m starts running with speed u (wrt planck). Find recoil velocity of planck.

$\vec{V}_{BP} = \vec{V}_B - \vec{V}_P$

$\vec{V}_B = \vec{V}_{BP} + \vec{V}_P$

$\vec{V}_B = u + (-v)$

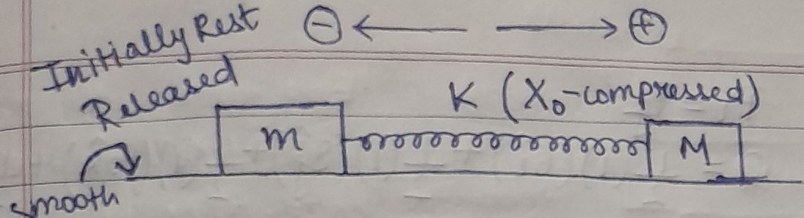
$= u - v$

Using M/C ⇒ $P_i = P_f$

$0 + 0 = -Mv + m(u-v)$

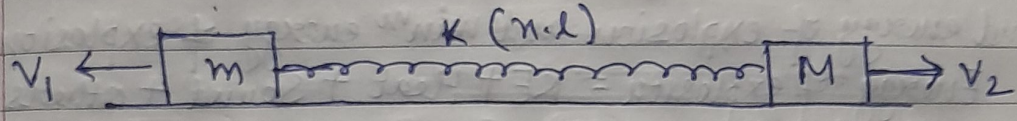
$\boxed{v = \frac{mu}{M+m}}$

Q Initially Rest Released



Find the max^m speed of blocks.

Max^m speeds will occure at natural length.



Using M/C ⇒ $P_i = P_f$

$0+0 = -mV_1 + MV_2$

$mV_1 = MV_2$ —①

Using TMEC ⇒ $K_i + U_i = K_f + U_f$

$= 0+0+\frac{1}{2}KX_0^2 = \frac{1}{2}mV_1^2 + \frac{1}{2}MV_2^2 + 0$

$= KX_0^2 = mV_1^2 + MV_2^2$ —②

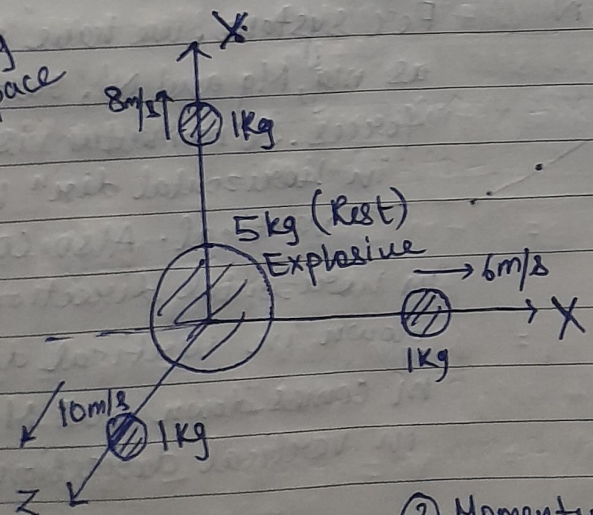
Using value of V_2 from ①

$$KX_0^2 = mV_1^2 + M\left(\frac{mV_1}{M}\right)^2$$

$$= KX_0^2 = mV_1^2 + \frac{m^2V_1^2}{M} \Rightarrow \boxed{V_1 = \sqrt{\frac{KMX_0^2}{(Mm+m^2)}}}$$

$$\boxed{V_2 = \sqrt{\frac{KmX_0^2}{(Mm+M^2)}}}$$

Q Gravity Free space



This explosive disintegrates into 4 parts in which 3 parts each of mass 1kg are moving as shown. Find velocity of 4th part.

Concept: ① COM of bomb will remain fixed even after explosion. (As $U_{com} = 0$, $F_{ext} = 0 \Rightarrow a_{com} = 0$)

② Momentum is conserved as there is no F_{ext}, only internal chemical energy are responsible.

Using M/C $\Rightarrow$ $P_i = P_f$

$$0 = (1)(6\hat{i}) + (1)(8\hat{j}) + (1)(10\hat{k}) + (2)(\vec{v})$$

$$\boxed{\vec{v} = -3\hat{i} - 4\hat{j} - 5\hat{k}} \qquad \boxed{v = \sqrt{3^2+4^2+5^2} = 5\sqrt{2}\ m/s}$$

(1) Find energy of explosion and min^m energy of explosion.

Energy of explosion = KE + Heat + Sound + others

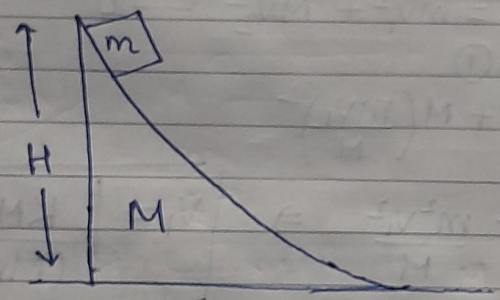
$$= \frac{1}{2}(1)(6)^2 + \frac{1}{2}(1)(8)^2 + \frac{1}{2}(1)(10)^2 + \frac{1}{2}(2)(5\sqrt{2})^2 + \text{Heat} + \text{Sound} + \text{others}$$

$$= 18 + 32 + 50 + 50 + \text{others}$$

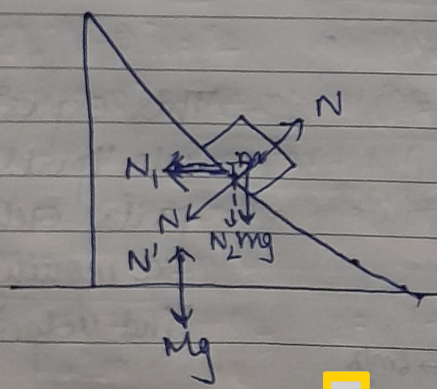
$$= 150 + \text{others J}$$

$\therefore$ Minm energy of explosion = 150 J

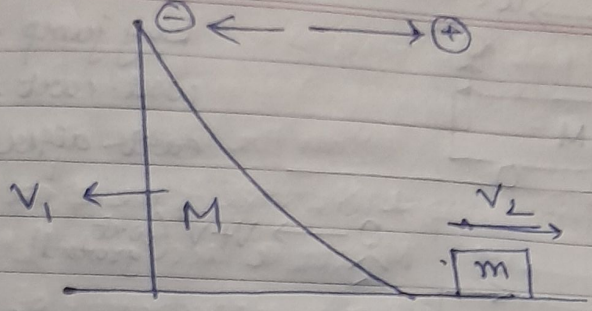
Q



Find speeds of blocks when m reaches ground.



Due to the horizontal component of $N \Rightarrow N_1$, M block moves left. For system, we have only forces as mg, Mg and N', rest are internal forces. So momentum is consumed in horizontal dirn but not in vertical. Also COM is fixed in horizontal direction but moves down in vertical direction as m comes down but M has no motion in vertical direction.



Using M/C in Horizontal:

$P_i = P_f$

$0 = -MV_1 + mV_2$

$MV_1 = mV_2$ — (1)

Using T.M.E.C ⇒ Loss in PE = Gain in KE

$$mgH = \frac{1}{2}MV_1^2 + \frac{1}{2}mV_2^2$$

$$2mgH = MV_1^2 + mV_2^2 \quad \text{— (2)}$$

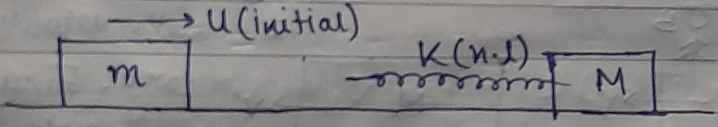
Using value of V_2 from (1)

$$2mgH = MV_1^2 + m\left(\frac{MV_1}{m}\right)^2$$

$$2mgH = MV_1^2 + \frac{M^2V_1^2}{m} \Rightarrow \boxed{V_1 = \sqrt{\frac{2m^2gH}{(Mm+M^2)}}}$$

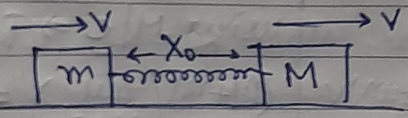
$$\boxed{V_2 = \sqrt{\frac{2MgH}{M+m}}}$$

Q



Find max^m compression in spring.

* Max^m compression will be when both blocks will have same velocity



Using M/C:

$P_i = P_f$

$mu + 0 = mV + MV$

$$\boxed{V = \frac{mu}{m+M}}$$

Using TMEC: $K_i + U_i = K_f + U_f$

$$= \frac{1}{2}mu^2 + 0 + 0 = \frac{1}{2}(m+M)V^2 + \frac{1}{2}KX_0^2$$

$$\Rightarrow mu^2 = (m+M)\left(\frac{mu}{m+M}\right)^2 + KX_0^2$$

$$KX_0^2 = mu^2 - \frac{m^2u^2}{m+M} = \frac{Mmu^2}{m+M}$$

$$\boxed{X_0 = u\sqrt{\frac{Mm}{K(m+M)}}}$$

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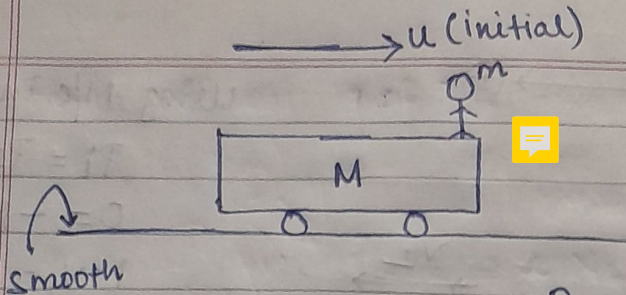
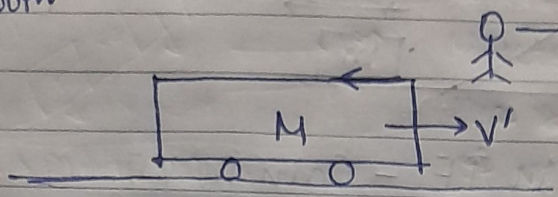


Boy jump with velocity v wrt. cart. Find velocity of cart after jump.



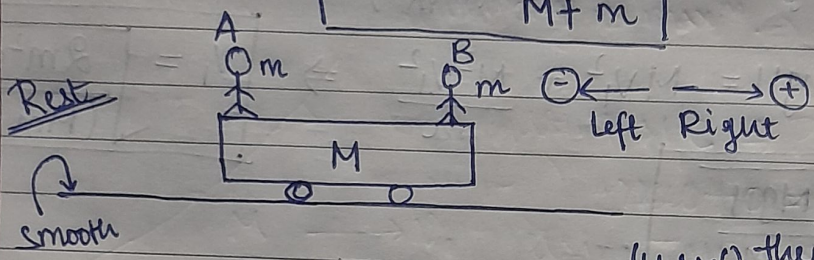
Using M/c along horizontal $\Rightarrow P_i = P_f$

$= (M+m)u = MV' + m(V+V')$

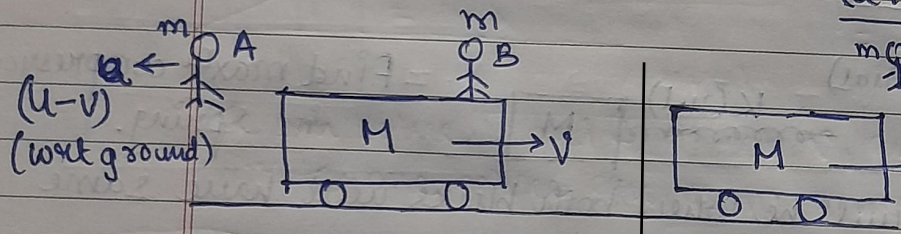
$= (M+m)u = (M+m)V' + mV$

$$V' = u - \frac{mv}{M+m}$$

Q Rest



Initially Boy A jump with velocity u (left) relative to cart, then cart starts moving right. Then Boy B jumps with velocity u (Right) wrt cart. Find final velocity of cart.



using M/C $\Rightarrow P_i = P_f$

$0 = m(u-v) + (M+m)V$

$mu - mv = MV + mV$

$mu = MV + 2mV$

$mu = (M+2m)V$

$$V = \frac{mu}{M+2m}$$

using M/C $\Rightarrow P_i = P_f$

$(M+m)\left(\frac{mu}{M+2m}\right) = MV' + m(u+v')$

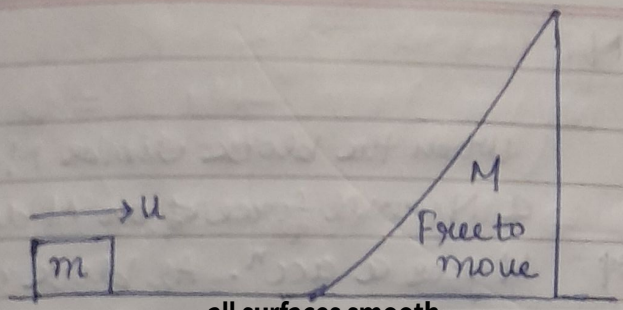
$= (M+m)\left(\frac{mu}{M+2m}\right) = MV' + mV' + mu$

$= (M+m)\left(\frac{mu}{M+2m}\right) = V'(M+m) + mu$

$= (M+m)\left(\frac{mu}{M+2m}\right) - mu = V'(M+m)$

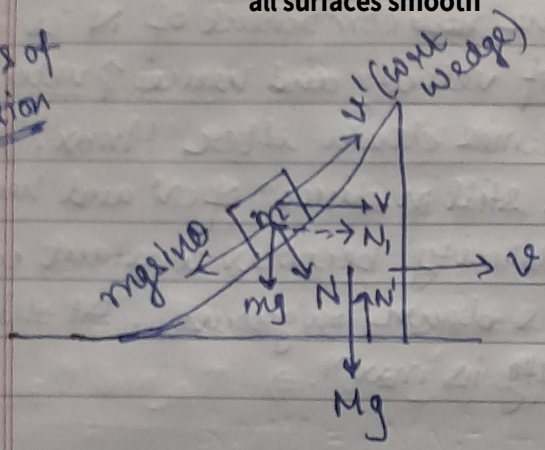
$$= V' = \frac{mu}{M+2m} - \frac{mu}{M+m}$$

Q



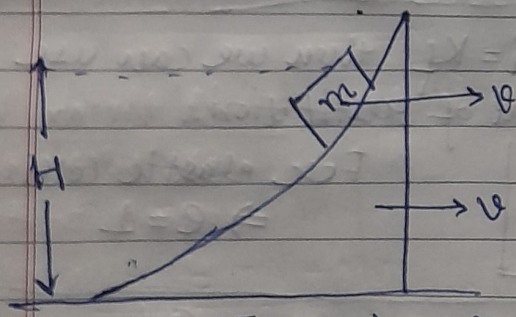
(a) Find max^m height achieved by m on M.

Analysis of situation



Let us assume that M has velocity v when block starts to climb it. By wedge constraint we know that m will have velocity only along common surface. Let m have velocity v' wrt wedge. Velocity of m wrt ground is resultant of v and v'. m applies a normal force on M whose component N_1 is responsible for velocity of M in horizontal dir^n. Component of mg along M ie mgsinθ is responsible for decrease in v'. When block reaches max height, v' must be zero. So m should now move with M with same velocity. No $\vec{F}_{ext}$ in hor^z ⇒ Momentum is conserved

* At max^m height both m and M have same velocity wrt ground.



Using M/C ⇒ $P_i = P_f$

$= mu + 0 = mv + Mv$

$= \boxed{v = \frac{mu}{M+m}}$

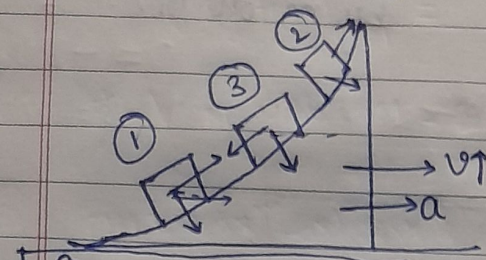
Using TMEC ⇒ $K_i + U_i = K_f + U_f$

$= \frac{1}{2}mu^2 + 0 = \frac{1}{2}(M+m)\left(\frac{mu}{M+m}\right)^2 + mgH$

$= H = \frac{u^2}{2g}\left(1 - \frac{m}{M+m}\right)$

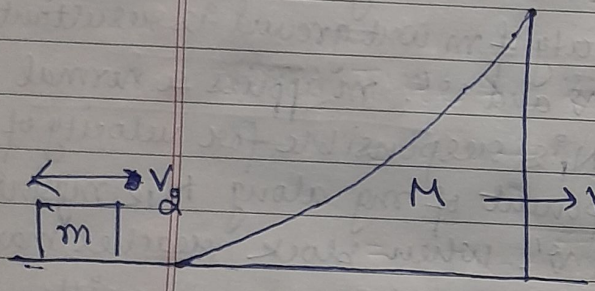
$$\boxed{H = \frac{Mu^2}{(M+m)2g}}$$

(b) Find max^m velocity of M.



① → Block climbing up
② → Block climbing up
③ → Block coming down

When the block climbs M, it applies a Normal force on M due to which M gets a acc^n. As block climbs up the component of N is in direc^n of velocity and hence ↑ velocity. When block comes down after Hmax, the comp. of N is still in direc^n of v and hence v still ↑. Velocity becomes constant when m comes down on ground. At this time the velocity is max^m.



Using M/C $\Rightarrow P_i = P_f$

$$mu + 0 = -mv_2 + Mv_1 \quad \text{—①}$$

Using TMEC $\Rightarrow K_i + U_i = K_f + U_f$

$$\frac{1}{2}mu^2 = \frac{1}{2}mv_2^2 + \frac{1}{2}Mv_1^2 \quad \text{—②}$$

We can solve ① and ② to get v_1 and v_2 but we can also write one more eqn using concept of collisions. (In detail later)

Concept

If $P_i = P_f$ and also $K_i = K_f$ then we can use concept of collisions. e = coefficient of restitution.

$$e = \frac{\text{Velocity of sep}}{\text{Velocity of app.}}$$

For elastic collisions $\Rightarrow e = 1$

So, $e = 1 = \frac{v_1 + v_2}{u} \Rightarrow v_1 + v_2 = u \quad \text{—②'}$

$\Rightarrow v_2 = u - v_1$

Using ① and ②'

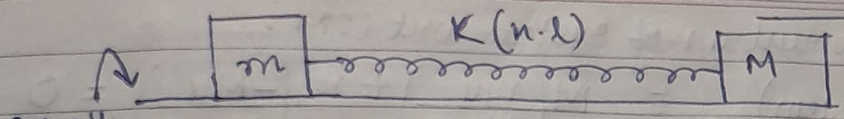
$mu = -mu + mv_1 + Mv_1$

$2mu = v_1(m + M)$

$$\boxed{v_1 = \frac{2mu}{M+m}} \rightarrow \text{Max}^m \text{ velocity of M}$$

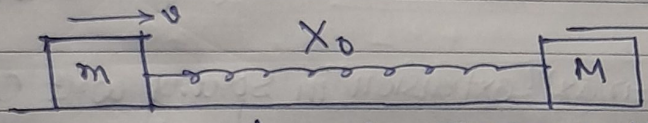
$v_2 = u - v_1$

$$\boxed{v_2 = \frac{(M-m)u}{M+m}}$$

Q 

① Find max^m ext in spring.

* Max^m extension will be when both blocks have same velocity.



Using M/C:

$P_i = P_f$

$0 + Mu = mv + Mv$

$$v = \frac{Mu}{m+M}$$

Using THEC $\Rightarrow K_i + U_i = K_f + U_f$

$= 0 + \frac{1}{2}Mu^2 + 0 = \frac{1}{2}(m+M)v^2 + \frac{1}{2}KX_0^2$

$= Mu^2 = (m+M)\frac{M^2u^2}{(m+M)^2} + KX_0^2$

$= KX_0^2 = Mu^2 - \frac{M^2u^2}{m+M}$

$= KX_0^2 = \frac{Mmu^2 + M^2u^2 - M^2u^2}{M+m}$

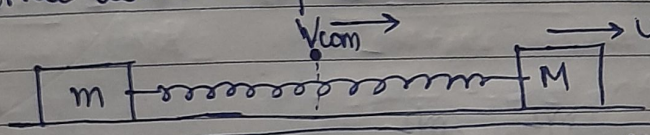
$$X_0 = u\sqrt{\frac{Mm}{K(M+m)}}$$

② Find velocity of COM.

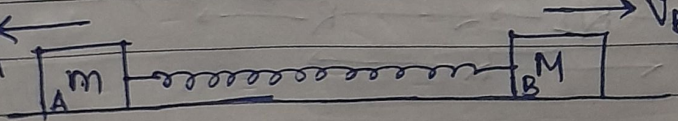
As F_{ext} on system $= 0 \Rightarrow P = \text{const} \Rightarrow V_{COM} = \text{const}$

$$V_{com} = \frac{m(0) + M(u)}{m+M} = \frac{Mu}{M+m}$$

③ Find initial velocities of blocks wrt COM.



COM ke respect me velocity nikalne ke liye COM ki velocity ko ulta krke jod do.

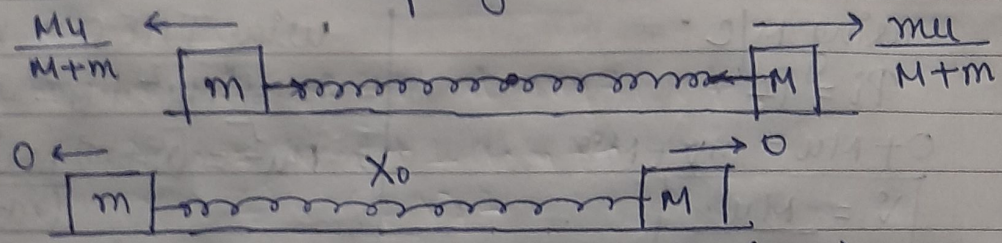


(a) Find momentum of system wrt COM.

$$(P_{system})_{COM} = M\left(\frac{mu}{M+m}\right) + m\left(\frac{-Mu}{M+m}\right) = 0$$

* V_{COM} wrt COM $= 0 \Rightarrow (P_{system})_{COM} = 0$. $(P_{system} = M\vec{V}_{com})$

(b) Find max^m extension in spring wrt COM.



→ Final velocities of block should be zero as $(P_{sys})_i = 0$ so to conserve momentum $(P_{sys})_f = 0 \Rightarrow v_{Ac}$ and $v_{BC} = 0$

Using TMEC ⇒ $K_i + U_i = K_f + U_f$

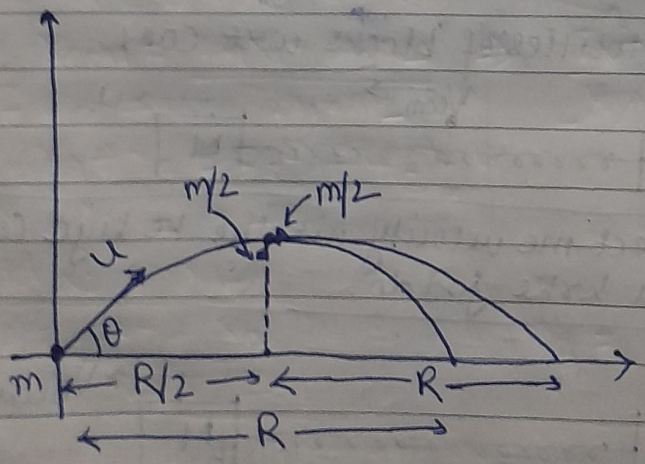
$$= \frac{1}{2}m\left(\frac{M^2u^2}{(M+m)^2}\right) + \frac{1}{2}M\left(\frac{m^2u^2}{(M+m)^2}\right) + 0 = 0 + 0 + \frac{1}{2}kX_0^2$$

$$= \boxed{X_0 = u\sqrt{\frac{mM}{k(M+m)}}}$$

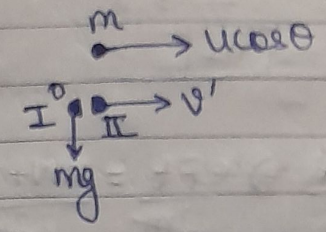
$$= \boxed{X_0 = u\sqrt{\frac{\mu}{k}}}$$

Note: $\frac{mM}{M+m} = \mu$ ↓ Reduced mass

Q



At Hmax ball explodes into 2 equal parts each of mass m/2. One part comes instantaneously at rest.
Find range of both parts.



To find v' we can use M/C

$P_i = P_f$

$m(u\cos\theta) = \frac{m}{2}(0) + \frac{m}{2}(v')$

$\boxed{v' = 2u\cos\theta}$

$\therefore$ Part I would fall under g from Hmax $\Rightarrow$ Range $= R/2$

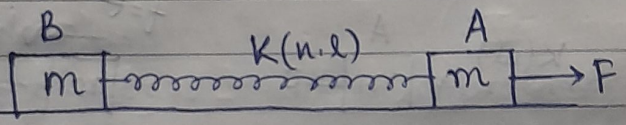
$\therefore$ Part II would now move with $v' = 2u\cos\theta$ that is double of the original velocity $\Rightarrow$ Range will also be double than original $\Rightarrow$ Range from Hmax $= R$

Range from origin $= \frac{R}{2} + R = \frac{3R}{2}$

** * Note: If momentum is conserved $\Rightarrow \vec{V}_{com} = \text{const}$

So, COM will travel the same path as it was travelling before explosion. $\Rightarrow$ Range of COM from origin $= R$

Q Initially Rest / At t=0



① Find acceleration of COM.

$a_{com} = \frac{\vec{F}_{ext}}{M} = \frac{F}{2m}$ $\quad\therefore\quad \boxed{a_{com} = \frac{F}{2m}}$

② Find velocity of COM at any time t.

$v_{com} = \cancel{u_{com}}^{0} + a_{com}t$

$\boxed{v_{com} = \frac{Ft}{2m}}$

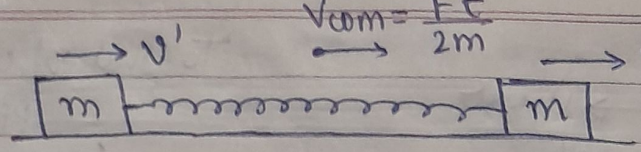
③ Find momentum of system at any time t.

$P_{sys} = M\vec{V}_{com} = 2m\left(\frac{Ft}{2m}\right)$

$\boxed{P_{sys} = Ft}$

④ At any time t, block A has velocity v. Find velocity of block B.

At $t=t$



$$V_{com} = \frac{m_1v_1 + m_2v_2}{m_1+m_2} = \frac{mv + mv'}{2m} \Rightarrow \frac{Ft}{2m} = \frac{mv+mv'}{2m}$$

$$\boxed{v' = \frac{Ft}{m} - v}$$

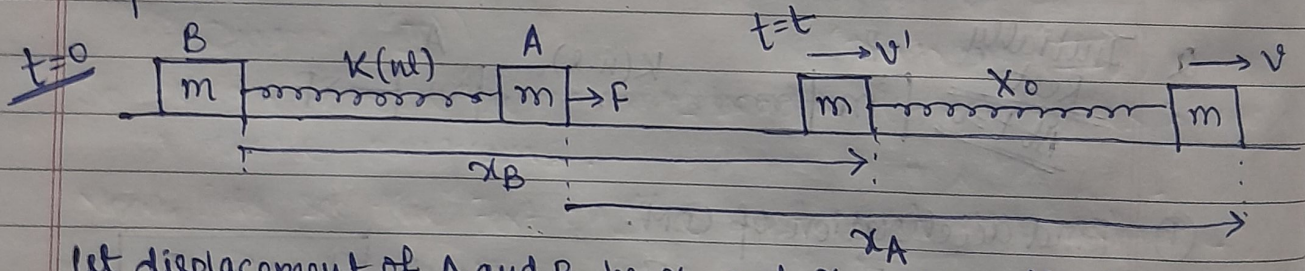
(5) Find displacement of COM at any time t.

$$x_{com} = \cancel{v_{com}^{0}t} + \frac{1}{2}a_{com}t^2$$

$$x_{com} = \frac{1}{2}\left(\frac{F}{2m}\right)t^2$$

$$\boxed{x_{com} = \frac{Ft^2}{4m}}$$

(6) At any time t, if extension in spring is x_0, then find displacement of each block.



Let displacement of A and B be x_A and x_B respectively.

$$x_A - x_B = x_0 \quad \text{—(1)}$$

Also, $x_{COM} = \frac{m_1x_1 + m_2x_2}{m_1+m_2} \Rightarrow \frac{Ft^2}{2\cancel{4}m} = \frac{mx_A + mx_B}{2m}$

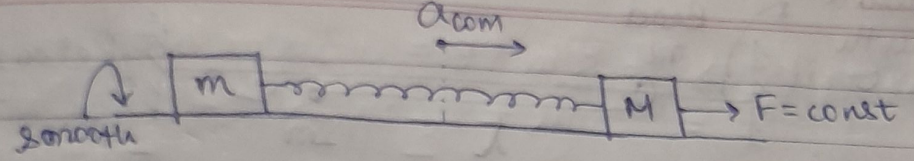
$$x_A + x_B = \frac{Ft^2}{2m} \quad \text{—(2)}$$

Add (1) and (2)

$$\boxed{x_A = \frac{Ft^2}{4m} + \frac{x_0}{2}}$$

$$\boxed{x_B = \frac{Ft^2}{4m} - \frac{x_0}{2}}$$

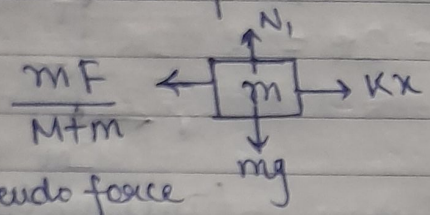
Q



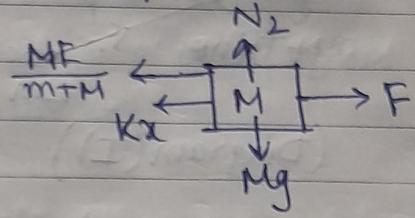
① Find acc^n of COM.

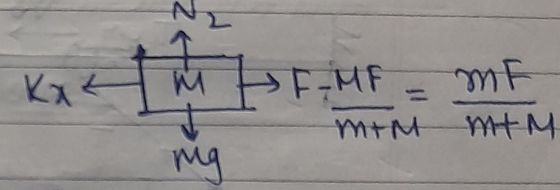
$$a_{com} = \frac{F_{ext}}{M_{sys}} = \frac{F}{m+M} \qquad \therefore \boxed{a_{com} = \frac{F}{m+M}}$$

② Draw FBD of each block wrt COM.



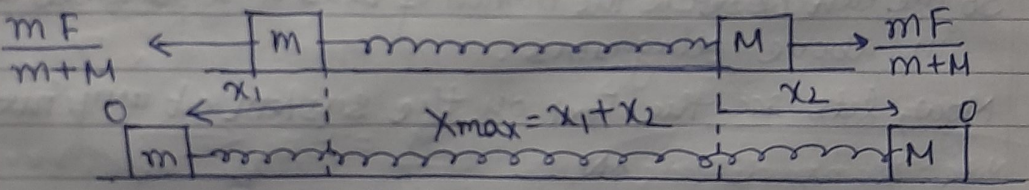
(Pseudo force as COM is non-inertial frame)





★ Blocks will perform oscillatory motion wrt COM.

③ Find max^m extension in spring.



We cannot use TMEC as ext Forces have work done.

Using WET ⇒ $W_{all} = \Delta KE$

$$\cancel{W_g}^{0} + \cancel{W_N}^{0} + W_s + W_F = \Delta KE$$

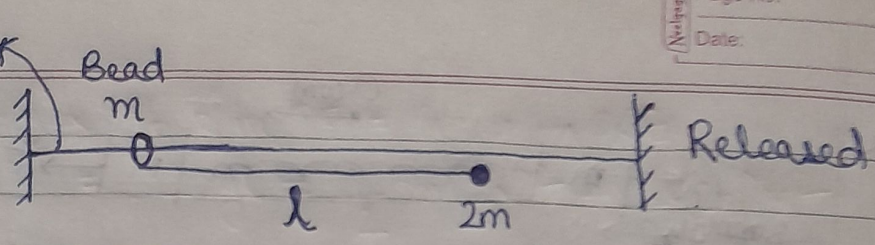
$$-\frac{1}{2}K(x_1+x_2)^2 + \frac{mF}{m+M}(x_1) + \frac{mF}{m+M}(x_2) = 0$$

$$\frac{1}{2}K(x_1+x_2)^2 = \frac{mF}{m+M}(\cancel{x_1+x_2})$$

$$x_1 + x_2 = \frac{2mF}{K(m+M)}$$

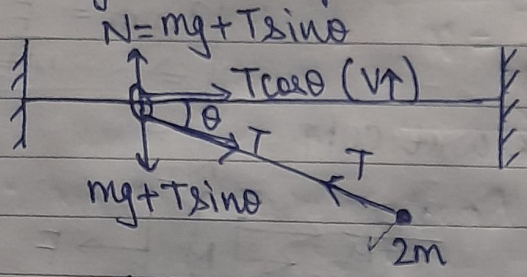
$$\boxed{x_{max} = \frac{2mF}{K(m+M)}}$$

Q

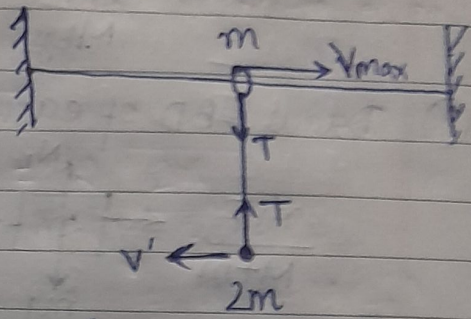


① Analyse motion of bead and particle

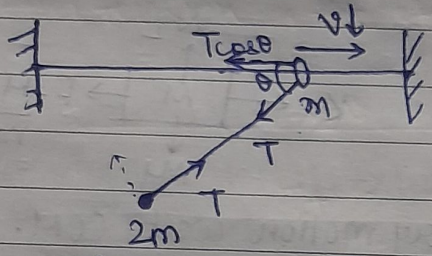
After time t



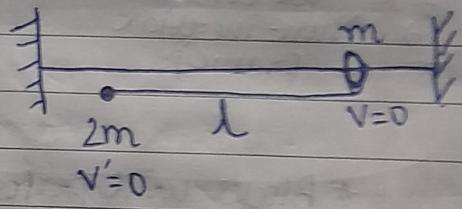
(Diagram (I))



(Diagram (II))



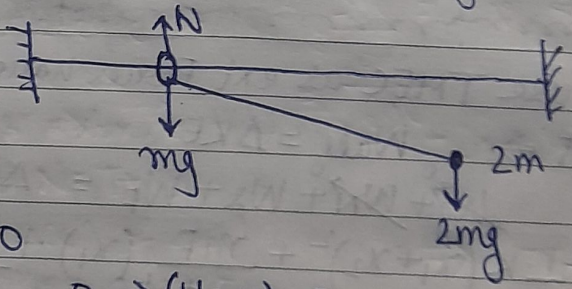
(Diagram (III))



(Diagram (IV))

So we can observe the motion of bead and particle as shown in above diagrams.

Now, lets see the Fext on the system at any instance

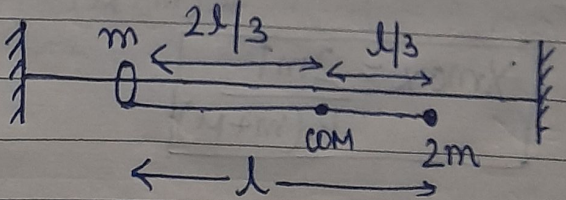


$\therefore F_{Hori} = 0$

$\Rightarrow (a_{com})_{Hori} = 0 \Rightarrow (u_{com})_H = 0$

$\therefore (x_{com})_H = 0$ (Displacement of COM)

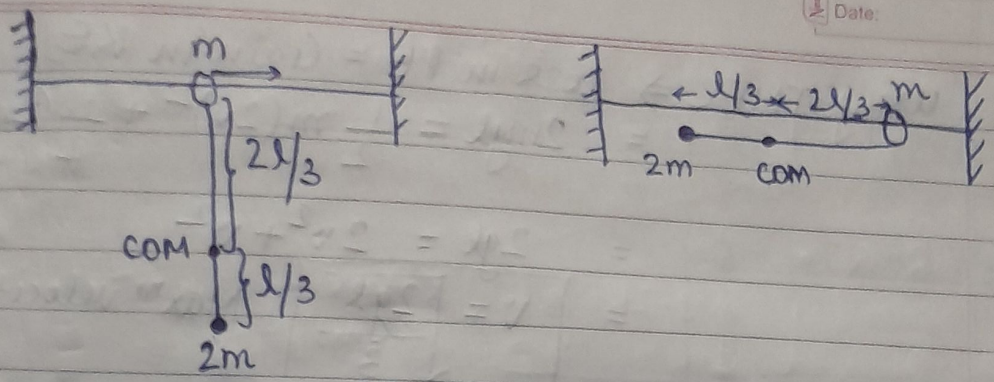
★ ∴ COM is moving only in vertical direction.



Position of COM from m

$= \frac{2m(l)}{m+2m} = \frac{2}{3}l$

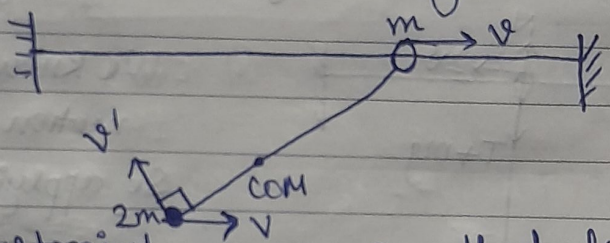
⇒ Position of COM from 2m $= \frac{l}{3}$



We can see that COM moves vertically and no motion of COM ● horizontally.

★ Also, bead is performing oscillatory motion of amplitude $A = \frac{2\ell}{3}$ about COM.

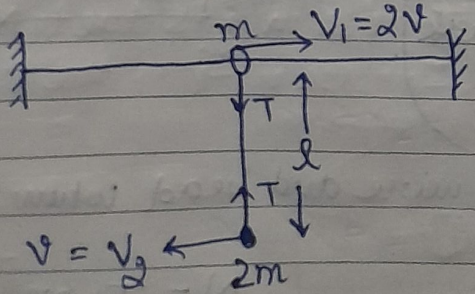
Now lets find how 2m is moving,



By string constraint we can say that length of string must be constant for which there should be no velocity along the line joining m and 2m. Velocity of 2m wrt m should be ⊥ to the line joining m and 2m. But velocity of 2m wrt ground should be vector sum v' and v (velocity of bead).

★ 2m particle is performing circular motion about mass m with velocity v'. Motion wrt ground is more complex.

(2) Find Vmax of Bead.



Using M/C in Horizontal:

$P_i = P_f$

$0 = mV_1 - 2mV_2$

$2mV_2 = mV_1$

$\boxed{V_1 = 2V_2}$

Using TMEC ⇒ Loss in PE = Gain in KE

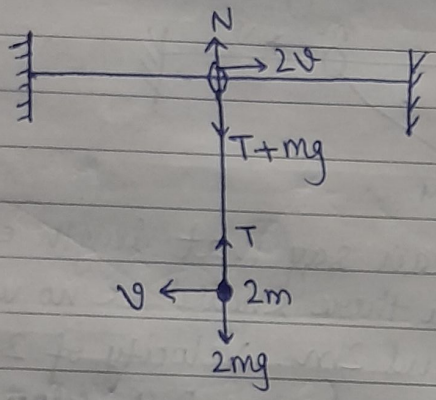
$$= 2mgl = \frac{1}{2}m(2v)^2 + \frac{1}{2}(2m)v^2$$

$$= 2gl = 2v^2 + v^2$$

$$= \boxed{v = \sqrt{\frac{2gl}{3}}}$$ Max^m velocity of particles

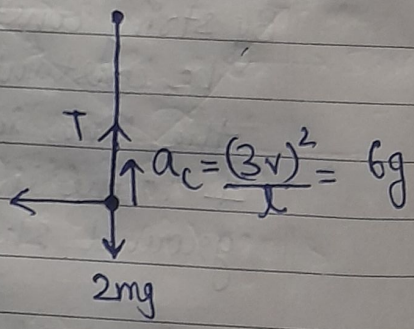
$\therefore$ Max^m velocity of bead $= 2v = 2\sqrt{\frac{2gl}{3}}$.

③ Find tension in the string when string becomes vertical.



We know 2m perform circular motion wrt m.

So applying NL on 2m wrt bead



$3\sqrt{\frac{2gl}{3}} = 3v$

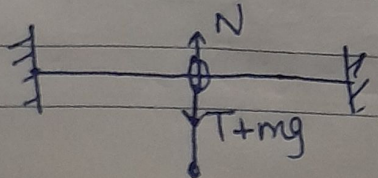
$= \sqrt{6gl}$

Note: No pseudo force in this situation as when string is vertical then for bead $N = T + mg$ and hence no acc^n of bead but for any other angle θ bead might have acc^n and then we will need to mark pseudo forces

$\Rightarrow T - 2mg = (2m)(6g)$

$\boxed{T = 14mg}$

④ Find normal force b/w wire and bead when string is vertical.



$N = T + mg = 14mg + mg$

$\boxed{N = 15mg}$

Impulse:

If $\vec{F}$ = const

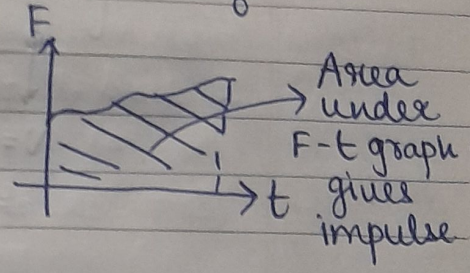
t = time interval for which $\vec{F}$ is acting.

Impulse $(\vec{J}) = \vec{F}t$

$$\therefore \boxed{\vec{J} = \vec{F}t}$$

If $\vec{F} \neq$ const

$$\vec{J} = \int d\vec{J} = \int_0^t \vec{F}\,dt$$



→ Unit = N-sec = kgm/s

→ Dimension = $[MLT^{-1}]$

→ Vector Quantity

→ Direction = ① Along $\vec{F}$ ② Along change in momentum.

Impulse-Momentum Theorem (IMT):

$$\vec{J} = \int \vec{F}\,dt$$

$$= \int m\frac{d\vec{v}}{dt}\,dt = m\int_{v_i}^{v_f} d\vec{v}$$

$$= m(\vec{v}_f - \vec{v}_i)$$

$$= m\vec{v}_f - m\vec{v}_i$$

$$\vec{J} = \vec{P}_f - \vec{P}_i$$

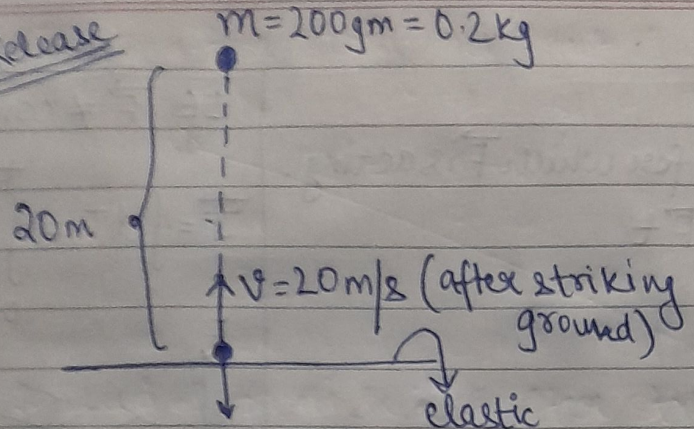
$$\therefore \boxed{\vec{J} = \vec{P}_f - \vec{P}_i = \Delta\vec{P}}$$

Also, $$\vec{J} = \int \vec{F}\,dt = \int_{P_i}^{P_f} \frac{d\vec{p}}{dt}\,dt = \vec{P}_f - \vec{P}_i$$

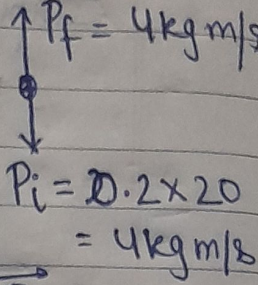
$$\therefore \boxed{\vec{J} = \vec{P}_f - \vec{P}_i} \qquad \boxed{\vec{J} = \Delta\vec{P}}$$

IMT states that net impulse is always equal to change in momentum.

Q Release



Find impulse on the ball when it strikes the floor.

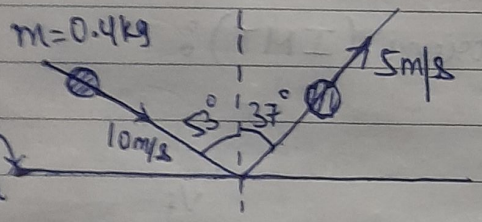


$\vec{J} = \vec{P_f} - \vec{P_i}$

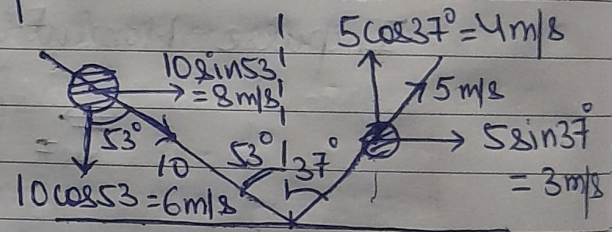
$= 4\hat{j} - 4(-\hat{j})$

$\boxed{\vec{J} = 8\hat{j}\ kg\,m/s}$

Q



① Find impulse applied by floor on the ball.



$P_i = 0.4kg(8\hat{i} - 6\hat{j})$

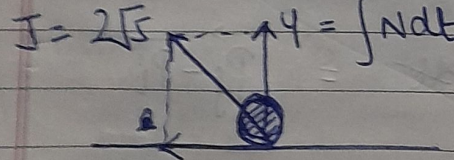
$= (3.2\hat{i} - 2.4\hat{j})\ kg\,m/s$

$P_f = 0.4kg(3\hat{i} + 4\hat{j})$

$= (1.2\hat{i} + 1.6\hat{j})\ kg\,m/s$

Using IMT $\Rightarrow \vec{J} = \vec{P_f} - \vec{P_i} = (-2\hat{i} + 4\hat{j})\ kg\,m/s$

② Find μ_k b/w ball and surface.



$J = \sqrt{2^2 + 4^2} = \sqrt{20} = 2\sqrt{5}\ kg\,m/s$

Impulse in $\hat{j}$ is due to Normal force.

Impulse in $-\hat{i}$ is due to friction.

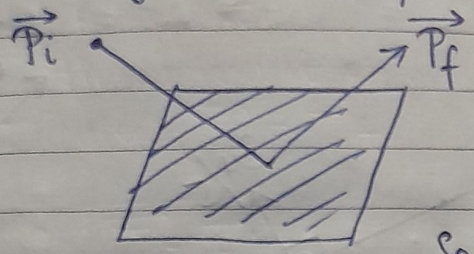
$\therefore \int N\,dt = 4 \quad -①$

$\int f_k\,dt = \mu_k \int N\,dt = 2 \quad -②$

Solving ① and ②

$\boxed{\mu_k = 0.5}$

Force by raindrops/hailstones on a surface:



Change in momentum of 1 Raindrop:

$= \vec{\Delta P} = \vec{P_f} - \vec{P_i}$

N Raindrops, $\vec{\Delta P_{net}} = N\vec{\Delta P}$

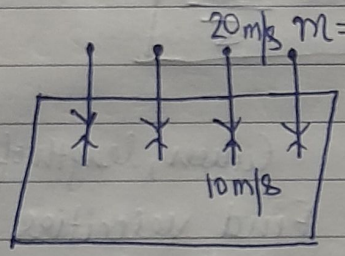
So, for time t: Force $= \frac{\vec{\Delta P_{net}}}{t} = \frac{N\vec{\Delta P}}{t}$

$$\boxed{\text{Force} = n\vec{\Delta P}}$$ (n = no. of raindrops striking per unit time)

∴ By Newtons 3rd law,

$$\boxed{\text{Force of Surface} = n\Delta P}$$

Q



10000 droplets are striking the surface per second. Find force on surface.

$n = 10000/\text{sec}$

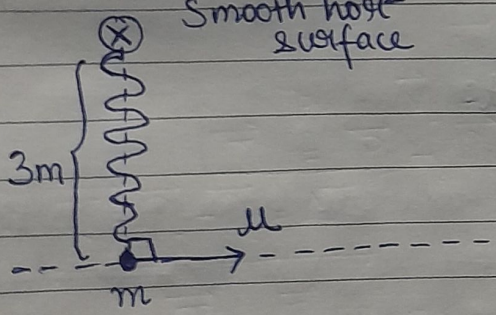
$\vec{P_i} = (2\times10^{-3}\times 20)(-\hat{j}) = 40\times10^{-3}(-\hat{j}) = 0.04(-\hat{j})$

$\vec{P_f} = (2\times10^{-3}\times10)\hat{j} = 20\times10^{-3}\hat{j} = 0.02\hat{j}$

$\vec{\Delta P} = \vec{P_f} - \vec{P_i} = 0.02\hat{j} - 0.04(-\hat{j}) = 0.06\hat{j}$ kg m/s

Force on surface $= n\Delta P = 10000\times0.06\,(-\hat{j}) = 600(-\hat{j})$ Newton

Q

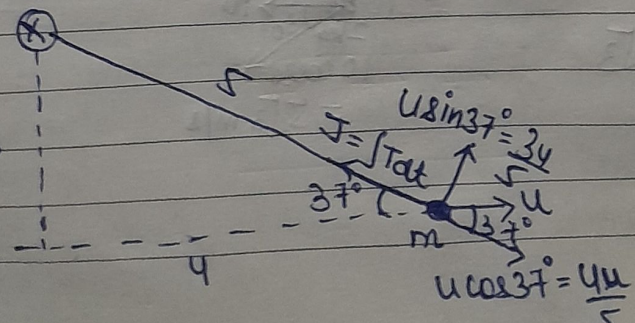


Given, length of slacked string = 5m

① Find speed of particle just after string get taut.

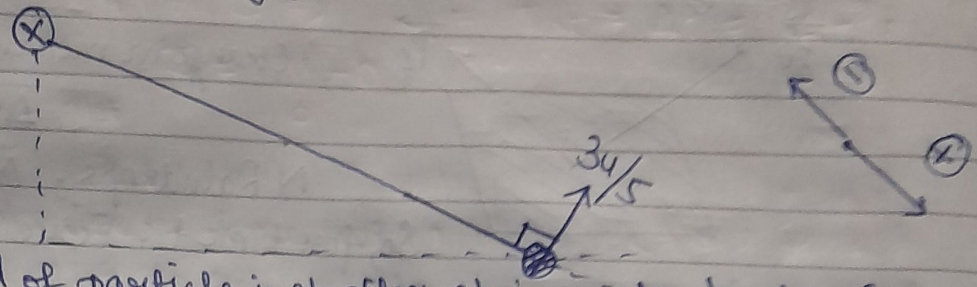
② Find impulse of tension when string just become taut.

Just before string get taut



Now, after string will get taut, there should be no vel. along line joining hinge and m acc to string constraint so impulse of tension should act opp to $\frac{4u}{5}$

Just after string get taut :

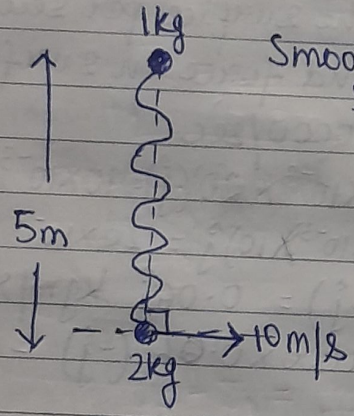


① ∴ speed of particle just after string gets taut $= \frac{3u}{5}$

② Impulse of tension : $-\vec{J} = -\int T dt = \vec{P_f} - \vec{P_i} = 0 - \frac{4mu}{5}$

$$\therefore \boxed{\vec{J} = \int T dt = \frac{4mu}{5}}$$

Q

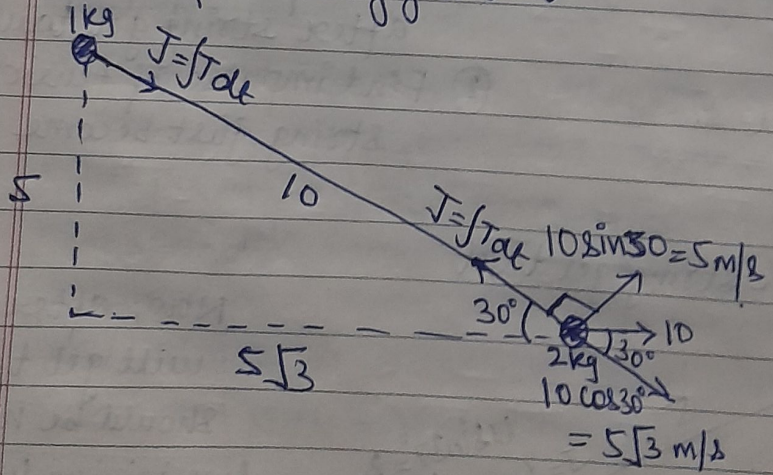


Given, length of string = 10m

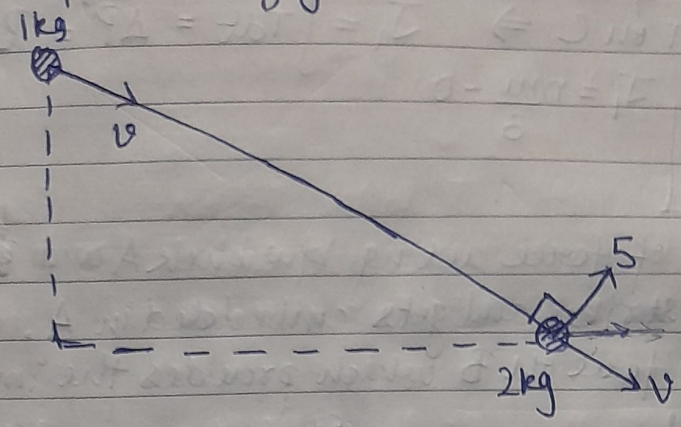
① Find velocities of particles just after string became taut.

② Find impulse of tension when string just became tight.

Just before string gets taut:



Just after string gets taut:



By string constraint there should be no relative motion along line joining particles, so both will have velocity v.

Applying M/C along string $\Rightarrow P_i = P_f$

$1\times0 + 2\times5\sqrt{3} = 1\times v + 2\times v$

$= 10\sqrt{3} = 3v$

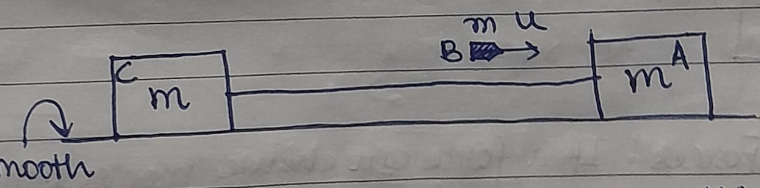
$= \boxed{v = \frac{10}{\sqrt{3}} \text{ m/s}}$

② Using IMT, $\vec{J} = \int T dt = \vec{P_f} - \vec{P_i}$ on 1kg

$\vec{J} = 1\times\frac{10}{\sqrt{3}} - 0$

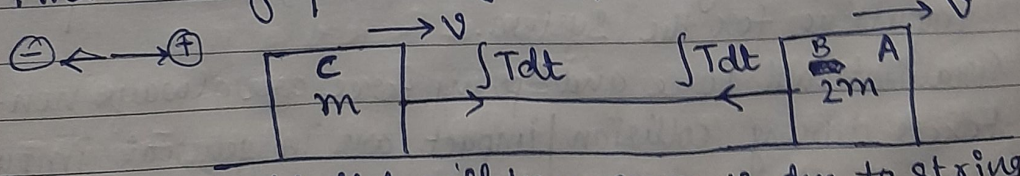
$\boxed{\vec{J} = \frac{10}{\sqrt{3}} \text{ kgm/s}}$

Q



Bullet B strikes block A and gets embedded in it.

① Find velocity of blocks just after collision.



* Both blocks and bullet will have same v due to string constraint.

Applying M/C on A+B+C $\Rightarrow P_i = P_f$

$0 + mu + 0 = 2mv + mv$

$mu = 3mv$

$\boxed{v = \frac{u}{3}}$

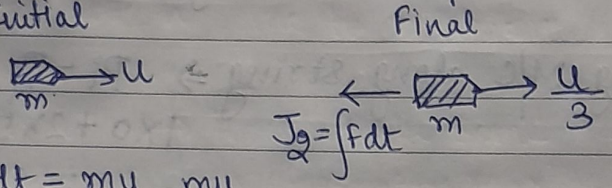
② Find impulse of tension.

Applying IMT on C ⇒ $J_1 = \int T dt = \Delta\vec{P} = P_f - P_i$

$J_1 = \frac{mu}{3} - 0$

$\therefore \boxed{J_1 = \frac{mu}{3}}$

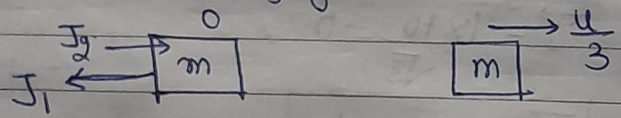
③ Find impulse of force acting b/w block A and ~~Block~~ bullet B. when bullet strikes and gets embedded in A, block A applies a resistive force on B which provides the impulse.



$-J_2 = -\int F dt = \frac{mu}{3} - mu$

$\therefore \boxed{J_2 = \frac{2mu}{3}}$

Alternate: Applying IMT on block A.



$J_2 - J_1 = \frac{mu}{3} - 0 \Rightarrow J_2 - \frac{mu}{3} = \frac{mu}{3}$

$\therefore \boxed{J_2 = \frac{2mu}{3}}$

Impulsive Forces: If a force can change momentum of a system in very small time interval, then this force is impulsive for the system.

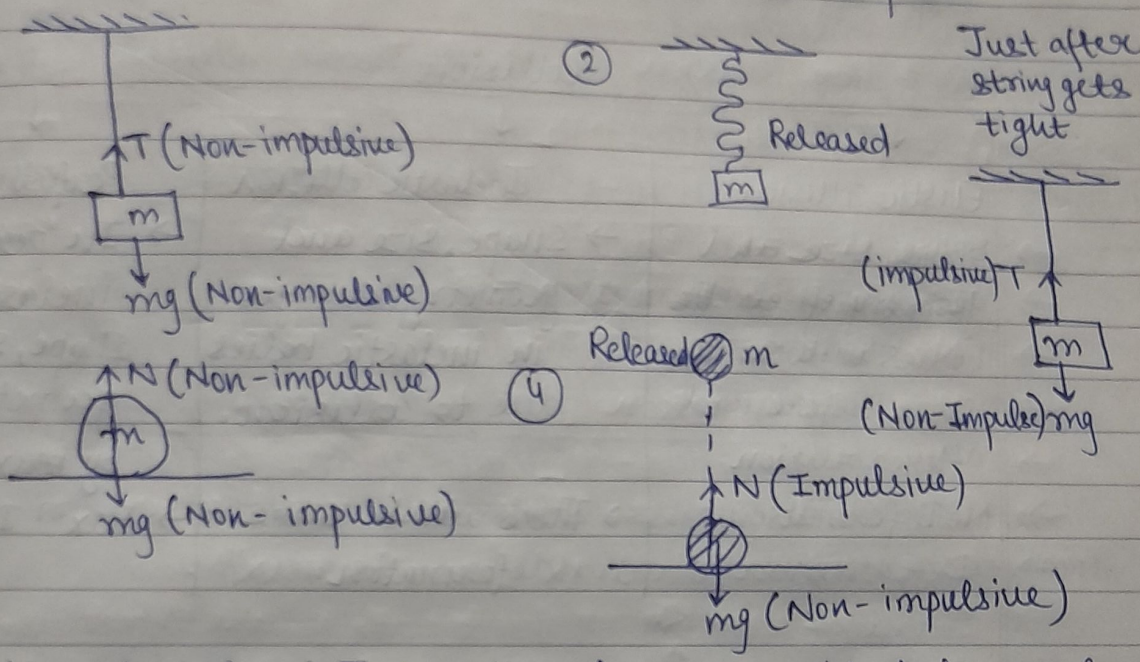
→ Gravitational Force and Spring Force are always non-impulsive.

→ Forces during collision/impact are in general impulsive.

→ Only impulsive force can balance other impulsive force.

Ex
② During
③ After collision

Q Judge whether forces are impulsive or non-impulsive.



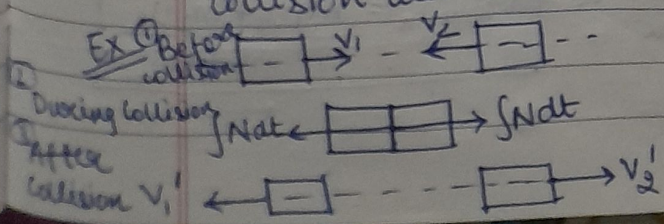
Line of Motion : It is a line along which object is moving or the line along which we have velocity

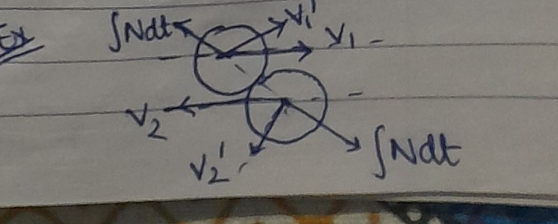
Line of Impact : It is a line along which interacting bodies are applying force on each other.

Collision : Redistribution of linear momentum among interacting bodies is known as collision.
In general, in collision, interacting bodies will apply impulsive forces on each other.

Collision

Head-on Collision	Oblique Collision
→ Line of motion and line of impact coincide.	→ Line of motion and line of impact are different.
→ Line of motion before and after collision will be same.	→ Line of motion before and after collision may be different.





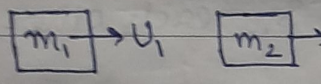
Types of collision based on material:

Collision

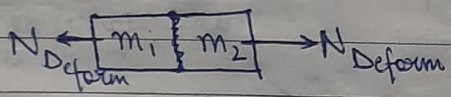
Elastic Collision	Inelastic Collision	Perfectly inelastic
→ Shape, size and geometry of elastic bodies will remain same before and after collision.	→ Shape, size and geometry changes in inelastic bodies due to collision.	→ In perfectly inelastic bodies, shape, size and geometry changes by max^m amount.
→ No Deformation	→ There will be some deformation.	→ Max^m Deformation

Process of collision:

① Before Collision ($u_1 > u_2$)

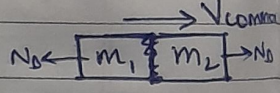


② During Collision



$u_1 \downarrow$ $u_2 \uparrow$

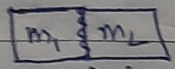
③ When both have same velocity



Max^m Deform

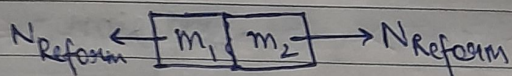
Process of Deformation

③ V_{common}

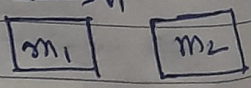


Max^M Deform

④ Reformation starts when $u_2 > u_1$



⑤ After Collision



Process of Reformation

Now, using M/C on system:

Before and after collision $\Rightarrow$ $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ — (1)

Before Collision upto deformation $\Rightarrow$ $m_1u_1 + m_2u_2 = (m_1 + m_2)v_{common}$

$\therefore$ $$v_{common} = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

Coefficient of Restitution = $\boxed{e = \frac{\text{Impulse of Reformation}}{\text{Impulse of Deformation}}}$

$e = \frac{\int N_R dt}{\int N_D dt} = \frac{m_2 v_2 - m_2 v_{common}}{m_2 v_{common} - m_2 u_2}$ (Using IMT on m_2 block)

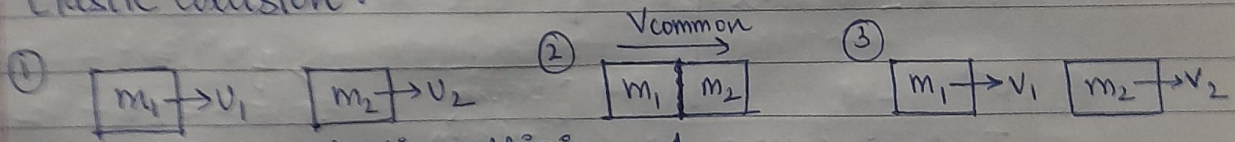
$= \frac{v_2 - \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}}{\frac{m_1 u_1 + m_2 u_2}{m_1 + m_2} - v_2} = \frac{m_1 v_2 + m_2 v_2 - m_1 v_1 - m_2 v_2}{m_1 u_1 + m_2 u_2 - m_1 v_2 - m_2 v_2}$

$e = \frac{m_1 v_2 + m_2 v_2 - (m_1 v_1 + m_2 v_2)}{m_1 u_1 - m_1 v_2}$ (using eqn ① $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$)

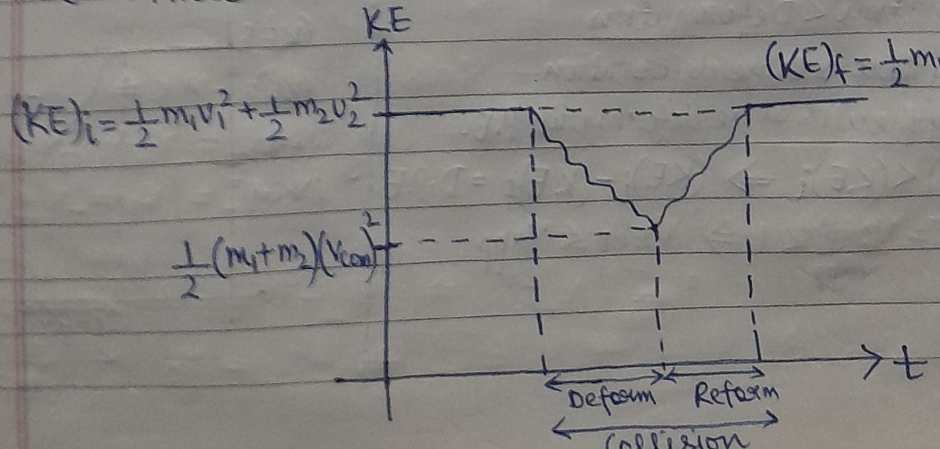
$e = \frac{m_1 v_2 - m_1 v_1}{m_1 u_1 - m_1 v_2} = \frac{v_2 - v_1}{u_1 - u_2} = \frac{\text{Velocity of separation}}{\text{Velocity of approach}}$

$\therefore \boxed{e = \frac{\text{Velocity of separation (After Collision)}}{\text{Velocity of Approach (Before Collision)}} = \frac{v_2 - v_1}{u_1 - u_2}}$

Elastic Collision:

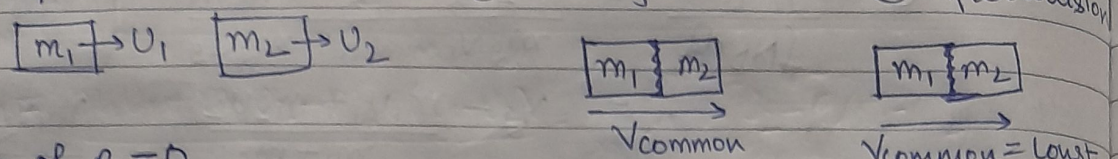


→ Value of e for elastic collision = 1
→ Impulse of Reformation = Impulse of Deformation
→ Finally blocks will achieve same shape, size and geometry as initial.
→ Finally No deformation
→ Velocity of sep. = Velocity app.
→ $(KE)_{Final} = (KE)_{initial}$

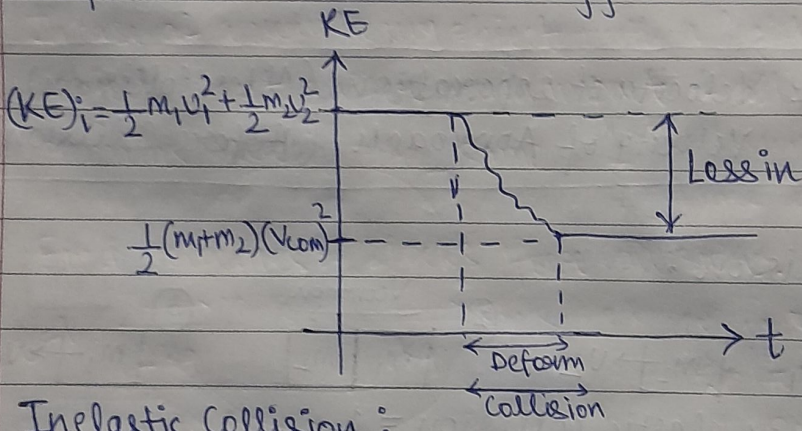


Perfectly Inelastic Collision:

① Before collision ② During collision ③ After Collision

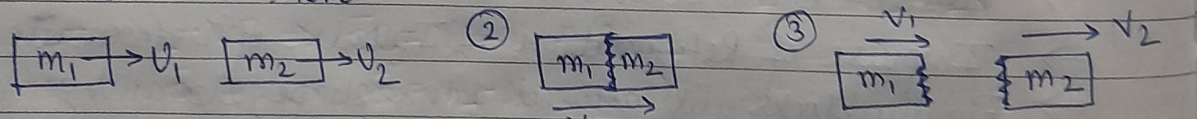


→ Value of $e = 0$
→ Impulsive Reformation = 0
→ No Reformation
→ Max^m Deformation
→ $v_{sep} = 0$
→ Both blocks strikes, sticks and move together
→ Deformation Potential Energy is max^m.

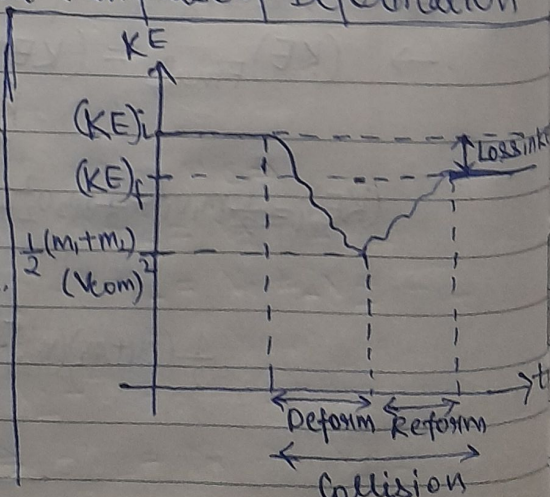


∴ Loss in KE = Deformation P.E.

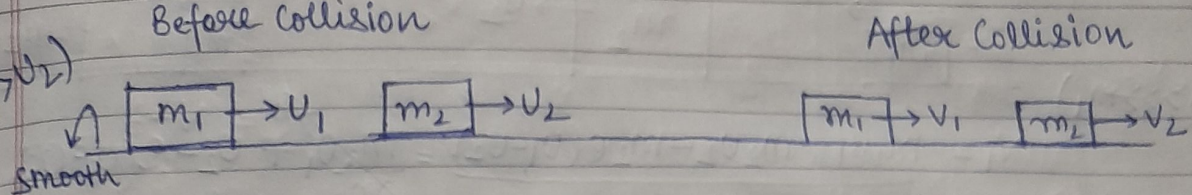
Inelastic Collision:



→ Value of e: $0 < e < 1$
→ $0 < e = \frac{\int N_R dt}{\int N_D dt} < 1 \Rightarrow$ Impulse of Reformation < Impulse of Deformation
→ Reformation < Deformation
→ Finally there is some deformation
→ Finally there is some deformation P.E.
→ $(KE_f) < (KE)_i \Rightarrow (KE)_i - (KE)_f =$ D.P.E.
→ $0 < e = \frac{v_{sep}}{v_{app}} < 1 \Rightarrow v_{sep} < v_{app}$



Head-on Collision:



Using M/C $\Rightarrow$ $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ —— (1)

Also, $e = \dfrac{v_2 - v_1}{u_1 - u_2}$ —— (2)

Now, put (2) in (1) $\Rightarrow$ $m_1u_1 + m_2u_2 = m_1v_1 + m_2(v_1 + e(u_1 - u_2))$

$\therefore$ $\boxed{v_1 = \dfrac{m_1u_1 + m_2u_2 - m_2e(u_1 - u_2)}{m_1 + m_2}}$

Put v_1 in (2) $\Rightarrow$ $v_2 = v_1 + e(u_1 - u_2)$

$= \dfrac{m_1u_1 + m_2u_2 - m_2e(u_1 - u_2) + e(u_1 - u_2)(m_1 + m_2)}{m_1 + m_2}$

$= \dfrac{m_1u_1 + m_2u_2 - m_2eu_1 + m_2eu_2 + m_1eu_1 - m_2eu_2 + m_2eu_1 - m_1eu_2}{m_1 + m_2}$

$\therefore$ $\boxed{v_2 = \dfrac{m_1u_1 + m_2u_2 + m_1e(u_1 - u_2)}{m_1 + m_2}}$

Special Cases:

① Case 1: Same mass elastic collision

$\Rightarrow m_1 = m_2 = m$ & $e = 1$

$v_1 = \dfrac{mu_1 + mu_2 - m(1)(u_1 - u_2)}{2m} = \dfrac{mu_1 + mu_2 - mu_1 + mu_2}{2m} = \dfrac{2u_2}{2} = u_2$ $\therefore \boxed{v_1 = u_2}$

$v_2 = \dfrac{mu_1 + mu_2 + m(1)(u_1 - u_2)}{2m} = \dfrac{u_1 + u_2 + u_1 - u_2}{2} = \dfrac{2u_1}{2} = u_1$ $\therefore \boxed{v_2 = u_1}$

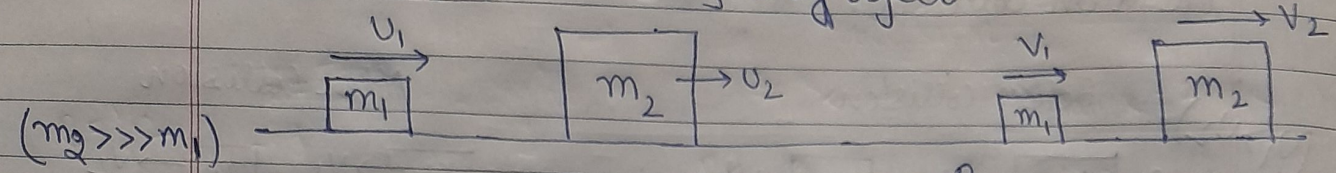
★ Velocities interchange in this case after collision.

② Case 2: Perfectly inelastic collision $(e = 0)$

$\boxed{v_1 = \dfrac{m_1u_1 + m_2u_2}{m_1 + m_2}}$ $\boxed{v_2 = \dfrac{m_1u_1 + m_2u_2}{m_1 + m_2}}$ $\therefore \boxed{v_1 = v_2}$

★ Both blocks move together with common velocity after collision

③ Case 3: Collision with very heavy object



$(m_2 >>> m_1)$

$$v_1 = \frac{m_1u_1 + m_2u_2 - m_2e(u_1-u_2)}{m_2} = \frac{m_1u_1}{m_2}^{\to 0} + u_2 - e(u_1-u_2)$$

$$\therefore \boxed{v_1 = u_2 - e(u_1-u_2)}$$

(Since $m_2 >>> m_1$, we can neglect m_1 in denom^r, also $\frac{m_1}{m_2} \approx 0$)

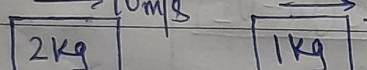
$$v_2 = \frac{m_1u_1 + m_2u_2 + m_1e(u_1-u_2)}{m_2} = \frac{m_1u_1}{m_2}^{\to 0} + u_2 + \frac{m_1e(u_1-u_2)}{m_2}^{\to 0}$$

$$\therefore \boxed{v_2 = u_2}$$

★ In case of a very heavy object with a light object, the heavier object wont suffer much impact of collision and its velocity remains nearly the same.

Q Find velocities of blocks after collision.

① $e=1$



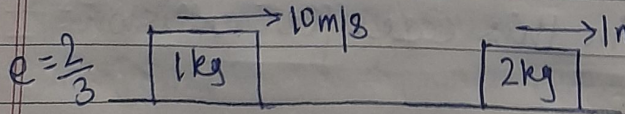
Using M/C ⇒
$20 + 5 = 2v_1 + v_2$
$2v_1 + v_2 = 25$ — (1)

$e = \frac{v_2 - v_1}{5} \Rightarrow 5e = v_2 - v_1$ — (2)
$5 = v_2 - v_1$ $(e=1)$

Put (2) in (1) ⇒ $2v_1 + 5 + v_1 = 25$
$3v_1 = 25 - 5 \Rightarrow \boxed{v_1 = 20/3 \text{ m/s}}$

$\boxed{v_2 = 5 + \frac{20}{3} = \frac{35}{3} \text{ m/s}}$

②



Using M/C ⇒
$10 + 2 = v_1 + 2v_2$
$12 = v_1 + 2v_2$ — (1)

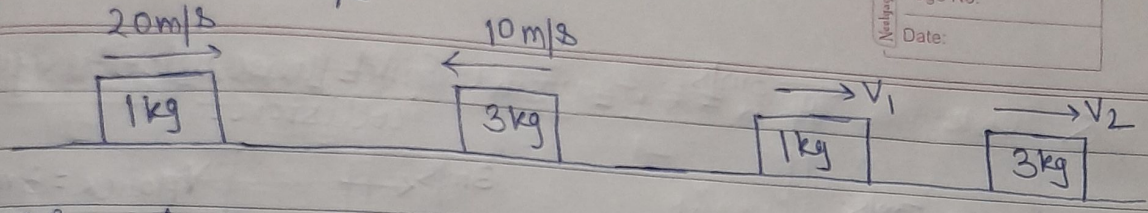
$e = \frac{v_2 - v_1}{9} = 9e = v_2 - v_1$
$9\left(\frac{2}{3}\right) = v_2 - v_1 \Rightarrow v_2 - v_1 = 6$ — (2)

Put (2) in (1) ⇒ $v_2 - 6 + 2v_2 = 12 \Rightarrow 3v_2 = 18 \Rightarrow \boxed{v_2 = 6 \text{ m/s}}$

$\boxed{v_1 = 0 \text{ m/s}}$

$e = 1/3$ $\quad \ominus \longleftrightarrow \oplus$

③



Using M/C ⇒ $20 - 30 = V_1 + 3V_2$

$= -10 = V_1 + 3V_2$ — ①

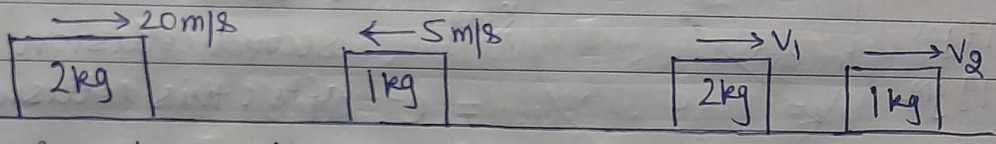
$e = \frac{V_2 - V_1}{20+10} \Rightarrow 30\left(\frac{1}{3}\right) = V_2 - V_1 \Rightarrow V_2 - V_1 = 10$ — ②

Put ② in ① ⇒ $-10 = V_2 - 10 + 3V_2$

$0 = 4V_2 \Rightarrow \boxed{V_2 = 0\ m/s}$

$\boxed{V_1 = -10\ m/s}$ (-ve means dirⁿ is opposite)

④ $e = \frac{3}{5}$



Using M/C ⇒ $40 - 5 = 2V_1 + V_2$

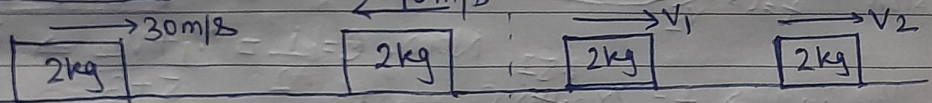
$35 = 2V_1 + V_2$ — ①

$e = \frac{V_2 - V_1}{25} \Rightarrow 25\left(\frac{3}{5}\right) = V_2 - V_1 \Rightarrow V_2 - V_1 = 15$ — ②

Put ② in ① ⇒ $35 = 2V_1 + 15 + V_1 \Rightarrow 20 = 3V_1 \Rightarrow \boxed{V_1 = \frac{20}{3}\ m/s}$

$\boxed{V_2 = \frac{65}{3}\ m/s}$

⑤ $e = 1/2$



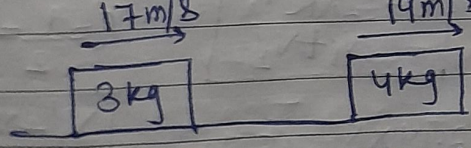
Using M/C ⇒ $60 - 20 = 2V_1 + 2V_2$

$20 = V_1 + V_2$ — ①

$e = \frac{V_2 - V_1}{40} = 40\left(\frac{1}{2}\right) = V_2 - V_1 \Rightarrow V_2 - V_1 = 20$ — ②

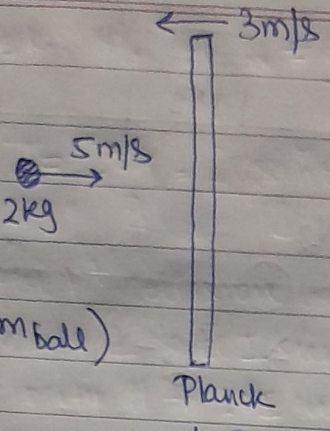
$\boxed{V_2 = 20\ m/s}$ $\quad \boxed{V_1 = 0\ m/s}$

⑥ $e = \frac{2}{7}$



Here $U_1 < U_2 \Rightarrow$ 3kg block wont be able to catch 4kg block so no collision occurs

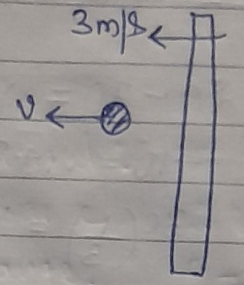
Q



$e = 1/2$

$(m_{planck} >> m_{ball})$

Find their velocities just after collision.



$V_{planck} = 3\,m/s$

$e = \frac{v-3}{8} = \frac{1}{2}$

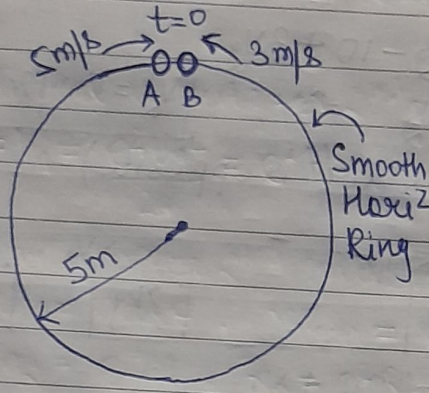
$V_{ball} = 7\,m/s$

Q

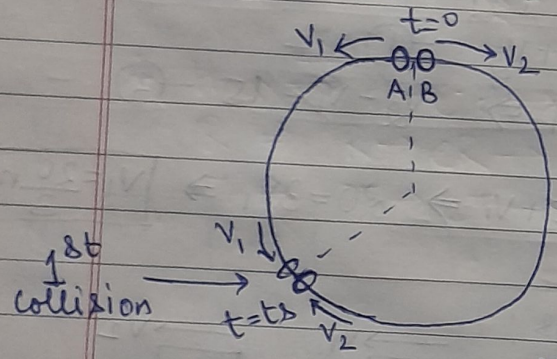
$e = 1/2$

$m_A = 17\,gm$

$m_B = 19\,gm$



Consider the collision at $t=0$ as zeroth collision. Find the time at which first and second collision happens after this collision.



1st collision ⇒

Distance covered by $A = V_1 t$

Distance covered by $B = V_2 t$

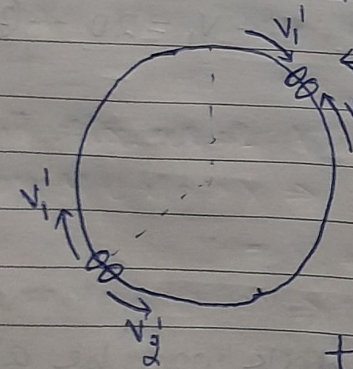
We know, $V_1 t + V_2 t = 2\pi R$

$$\Rightarrow \boxed{t = \frac{2\pi R}{V_1 + V_2}} \quad (1)$$

Now, $e = \frac{1}{2} = \frac{V_1 + V_2}{5+3} \Rightarrow \boxed{V_1 + V_2 = 4}$

Put in (1) $\Rightarrow t = \frac{2 \times 3.14 \times 5}{4} = 7.85\,s$

$\therefore$ 1st collision happens at 7.85s after 0th collision.



2nd collision ⇒

$V_1' t + V_2' t = 2\pi R$

$$\boxed{t = \frac{2\pi R}{V_1' + V_2'}}$$

$$e = \frac{1}{2} = \frac{V_1' + V_2'}{V_1 + V_2}$$

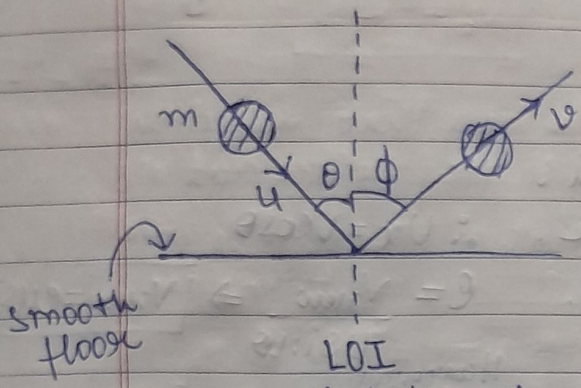
$$\boxed{V_1' + V_2' = 2}$$

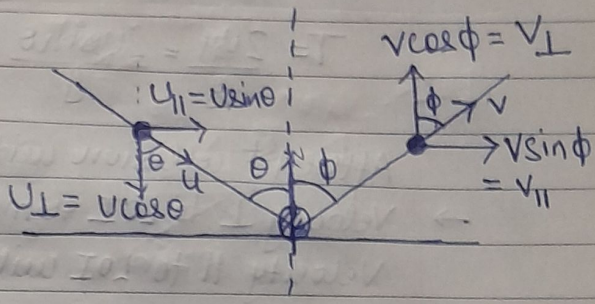
$$t = \frac{2 \times 3.14 \times 5}{2} = 15.7\,s$$

$\therefore$ 2nd collision happens at 15.7s after 1st collision.

Oblique Collision (2-D Collision):

⊥ → ⊥ to common surface
∥ → ∥ to common surface





When ball strikes the ground, it will experience an impulse by normal along the LOI, so velocity which is along the LOI will change but velocity ⊥ to LOI wont change as their is no force in that direction.

★ Velocity along common surface or velocity ⊥ to LOI will remain unchanged before and after collision.

$\therefore u_{\parallel} = v_{\parallel} \Rightarrow u\sin\theta = v\sin\phi$ — ①

★ $e = \dfrac{\text{Velocity of sep. along LOI}}{\text{Velocity of app. along LOI}}$

$\therefore e = \dfrac{v\cos\phi}{u\cos\theta}$ — ②

$\Rightarrow \boxed{v\cos\phi = eu\cos\theta = v_{\perp}}$

Now square and add ① and ②

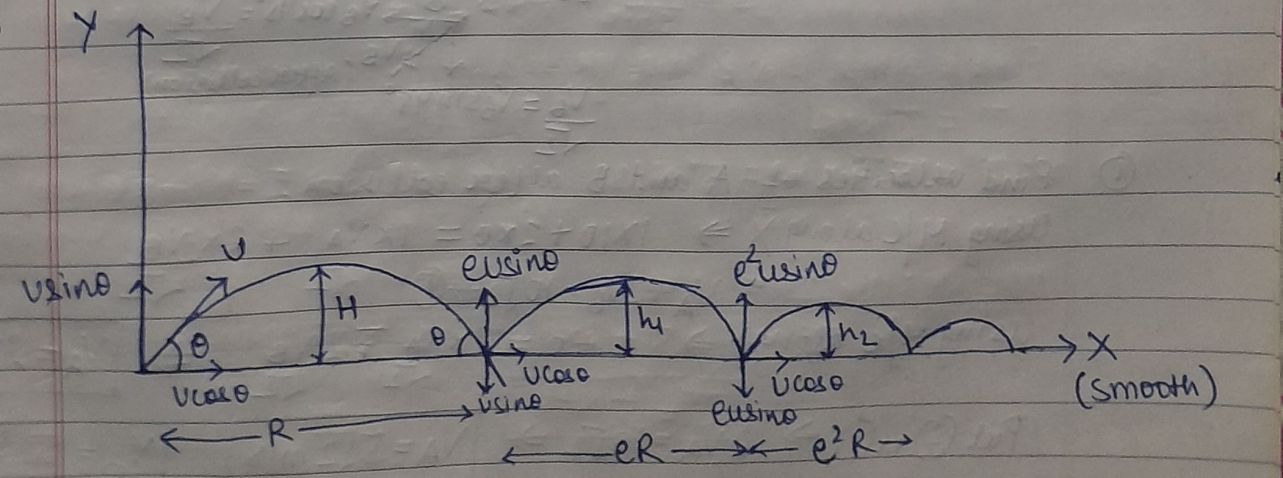
$v^2(\sin^2\phi + \cos^2\phi) = u^2(\sin^2\theta + e^2\cos^2\theta)$

$\therefore \boxed{v = u\sqrt{\sin^2\theta + e^2\cos^2\theta}}$

Also Divide ① and ②

$\boxed{\tan\phi = \dfrac{\tan\theta}{e}}$

**Ex



We know, $H_{max} = \frac{U_y^2}{2g} = \frac{U^2\sin^2\theta}{2g}$, $R = \frac{2U_xU_y}{g} = \frac{U^2\sin 2\theta}{g}$

$T = \frac{2U_y}{g} = \frac{2U\sin\theta}{g}$

After 1st collision with ground:

→ Velocity ⊥ to LOI remains same. $\therefore U_x = U\cos\theta$

Velocity ∥ to LOI will change. $e = \frac{V_{new}}{U\sin\theta} \Rightarrow \boxed{V_{new} = eU\sin\theta}$

$\therefore$ Initial velocity for 2nd projectile (∥ component) becomes $eU\sin\theta$.

$H_{max} = h_1 = \frac{e^2U_y^2}{2g} = e^2H$, $R_1 = \frac{2U_x(eU_y)}{g} = eR$, $T_1 = \frac{2(eU_y)}{g} = eT$

Similarly after 2nd collision with ground:

$U_x = U\cos\theta$, $V_{new} = e^2U\sin\theta$

$H_{max} = h_2 = e^4H$, $R_2 = e^2R$, $T_2 = e^2T$

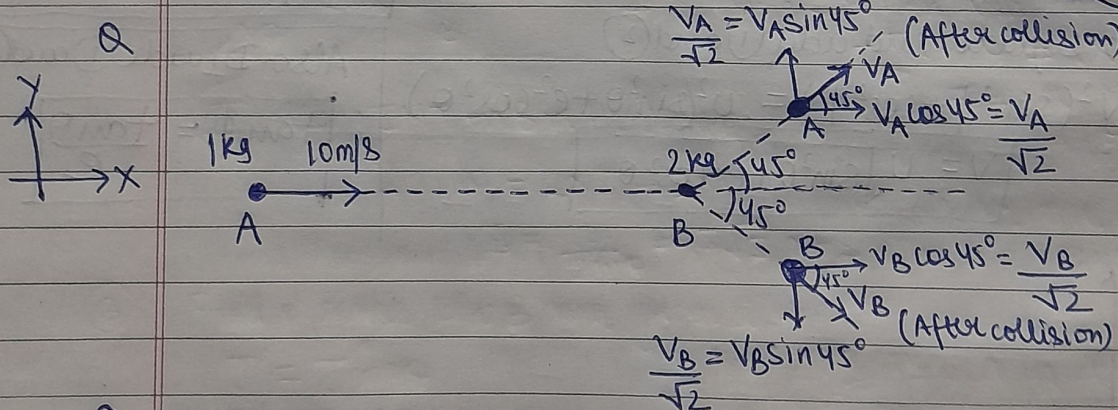
$\therefore$ $H_{max} \Rightarrow H \to e^2H \to e^4H \to e^6H \dots$

$R \Rightarrow R \to eR \to e^2R \to e^3R \dots$

$T \Rightarrow T \to eT \to e^2T \to e^3T \dots$

For inelastic collision $e<1$ So, H, R & T will keep decreasing after every collision

Q



① Find velocities of A and B after collision.

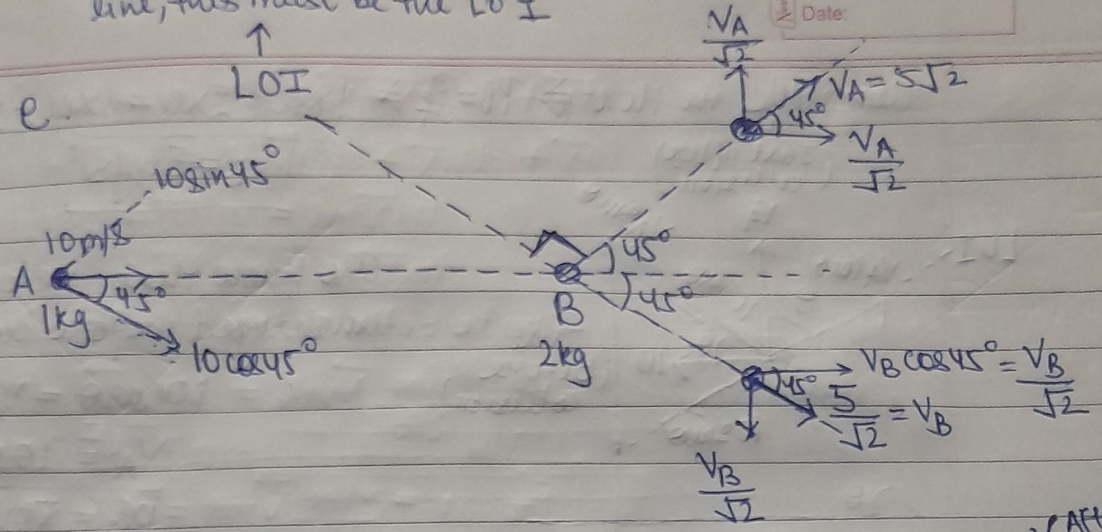
Using M/C along X ⇒ $1\times10 + 2\times0 = 1\times\frac{V_A}{\sqrt{2}} + 2\times\frac{V_B}{\sqrt{2}}$

$= V_A + 2V_B = 10\sqrt{2}$ — ①

Using M/C along Y ⇒ $0+0 = 1\times\frac{V_A}{\sqrt{2}} - 2\frac{V_B}{\sqrt{2}} \Rightarrow \boxed{V_A = 2V_B}$ — ②

Put ② in ① ⇒ $2V_A = 10\sqrt{2} \Rightarrow \boxed{V_A = 5\sqrt{2}\ m/s}$ $\boxed{V_B = \frac{5}{2}\sqrt{2}\ m/s}$

Since B was at rest initially and after collision it started moving along this line, this must be the LOI

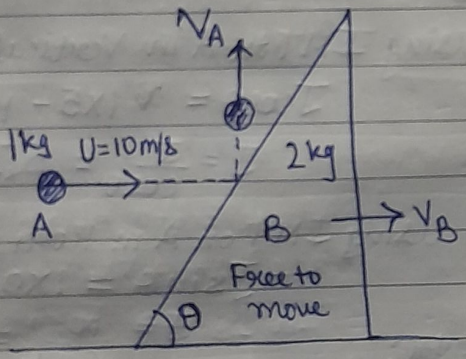


② Find e.

$$e = \frac{\text{Vel of sep along LOI}}{\text{Vel of sep along LOI}} = \frac{V_B(\text{along LOI}) + V_A(\text{along LOI})\ (\text{After collision})}{V_A(\text{along LOI}) + V_B(\text{along LOI})\ (\text{Before})}$$

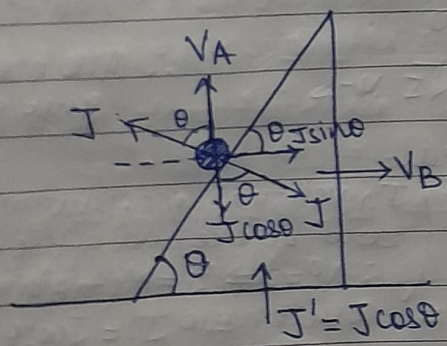
$$= \frac{\frac{5}{\sqrt{2}} + 0}{10\cos45° + 0} = \frac{5}{10} = \frac{1}{2} \quad \therefore \boxed{e = \frac{1}{2}}$$

(V_A has no component along LOI as it is $\perp$ to LOI (After collision))



Given, $\tan\theta = 2$

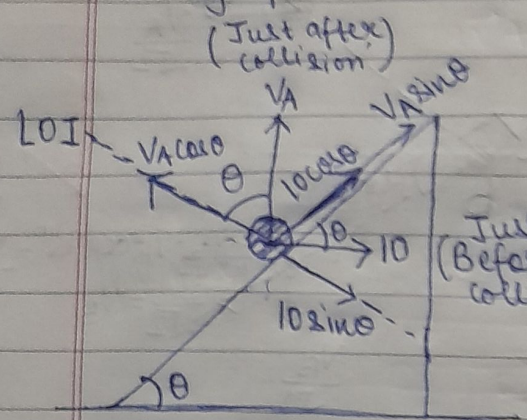
① Find velocity of block (V_B) and ball (V_A) just after collision.



When ball strikes the wedge it applies an impulse due to normal J to wedge's surface. By Newtons 3rd Law wedge applies same impulse on ball. Now component of J, $J\sin\theta$ is responsible for wedg's velocity V_B. $J\cos\theta$ will apply a force downwards, so to maintain equilibrium, ground will also apply an impulsive force $J' = J\cos\theta$. Also we can neglect any impulse by mg as it is non-impulsive. Now if we observe, we have only 3 Fext, ① Ball's mg ② Wedg's mg ③ J' all in vertical direction. (J is an internal force). So we have no ext. F on the system in Hori^z ⇒ Momentum is conserved in Hori^z.

Using M/C in Hor^{2} $\Rightarrow P_i = P_f \Rightarrow 1\times10 + 2\times0 = 1\times0 + 2V_B$

$$\boxed{V_B = 5\ m/s}$$



$\perp$ to LOI

We know that velocities remain unchanged before and after collision

$\therefore 10\cos\theta = V_A\sin\theta$

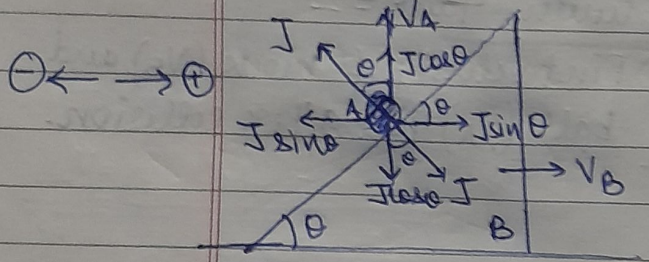
$V_A = 10\cot\theta$

$$\boxed{V_A = 5\ m/s}$$

$$\therefore \boxed{\begin{aligned} V_A &= 5\ m/s \\ V_B &= 5\ m/s \end{aligned}}$$

Note: mg is a non-impulsive force so its component $mg\sin\theta$ would take time to change vel. $\perp$ to LOI in this case. So just before and after collision velocities $\perp$ to LOI still remains unchanged

(2) Find impulse b/w A and B



(I) Using IMT on A in Vertical: $(J = \Delta P)$

$J\cos\theta = 1\times5 - 1\times0$

$J = \frac{5}{\cos\theta} = 5\sqrt{5}\ N\text{-}s$

(II) Using IMT on A in Horizontal:

$-J\sin\theta = 1\times0 - 1\times0$

$J = \frac{10}{2\sin\theta} = 5\sqrt{5}\ N\text{-}s$

(III) Using IMT on B in Horizontal:

$J\sin\theta = 2\times5 - 2\times0$

$J = \frac{10}{2\sin\theta} = 5\sqrt{5}\ N\text{-}s$

$\therefore$ All methods will give same answer

(3) Find impulse b/w B and ground.

$J' = J\cos\theta$

$J' = 5\sqrt{5}\times\frac{1}{\sqrt{5}} = 5\ N\text{-}s$

(In vertical momentum of B was zero before and after collision $\therefore \Delta P = 0 \Rightarrow J = 0$ (on B in vert). So we must have $J' = J\cos\theta$ to balance out the $J\cos\theta$ by normal)

Alternate: Use IMT on A+B in vertical:

$J' = (1\times5 + 2\times0) + (1\times0 + 2\times0)$

$\therefore \boxed{J' = 5\ N\text{-}s}$

④ Find e.

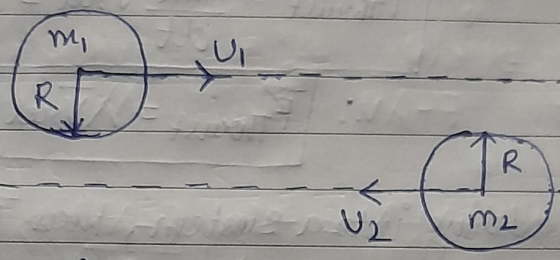
$$e = \frac{\text{Vsep along LOI (after collision)}}{\text{Vapp along LOI (before collision)}} = \frac{5\cos\theta + 5\sin\theta}{10\sin\theta + 0}$$

$$e = \frac{1}{2}\cot\theta + \frac{1}{2} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} \quad \therefore \boxed{e = \frac{3}{4}}$$

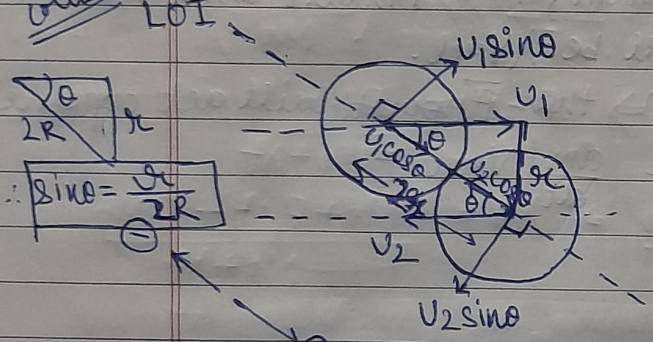
Oblique Collision between balls :

If $x = 2R$ ⇒ Balls just touch
If $x > 2R$ ⇒ Balls dont collide

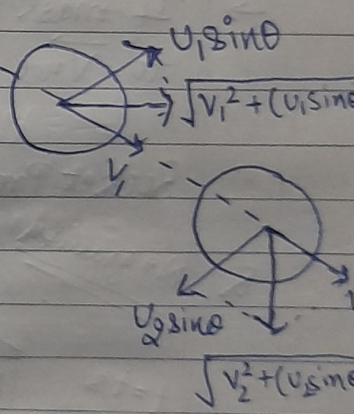
Before collision



During collision



After collision



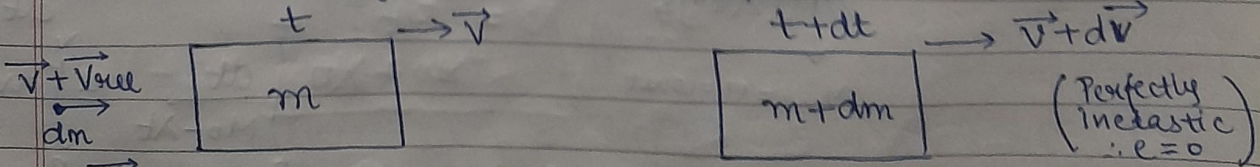
* Velocities ⊥ LOI remain unchanged before and after collision.

Using M/C along LOI ⇒ $m_1U_1\cos\theta - m_2U_2\cos\theta = m_1V_1 + m_2V_2$ — ①

$$e = \frac{V_2 - V_1}{U_1\cos\theta + U_2\cos\theta} \quad -②$$

∴ We can solve ① and ② to get V_1 and V_2

Variable Mass System (Thrust Force):



$\vec{V}_{rel}$ = Velocity of dm wrt m

∴ Velocity of dm wrt ground = $\vec{V} + \vec{V}_{rel}$

Using M/C ⇒ $m\vec{v} + dm(\vec{v}+\vec{v}_{rel}) = (m+dm)(\vec{v}+d\vec{v})$

$m\vec{v} + dm\vec{v} + dm\vec{v}_{rel} = m\vec{v} + m d\vec{v} + dm\vec{v} + dm\,d\vec{v}$

$dm\vec{v}_{rel} = m d\vec{v} + dm\,d\vec{v}$ (very small ≈ 0)

$dm\vec{v}_{rel} = m d\vec{v}$

Divide both sides by dt

$$\frac{dm}{dt}\vec{v}_{rel} = m\frac{d\vec{v}}{dt} \qquad \left(m\frac{d\vec{v}}{dt} = \vec{F}_{Thrust}\right)$$

$$\boxed{\vec{F}_{Thrust} = \frac{dm}{dt}\vec{v}_{rel} = \lambda\vec{v}_{rel}} \qquad \left(\lambda = \frac{dm}{dt}\right)$$

$$\therefore \boxed{\vec{F}_{Thrust} = \lambda\vec{v}_{rel}}$$

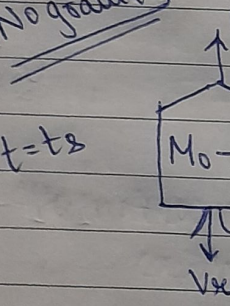
λ → kg/s → $[MT^{-1}]$

→ If mass is adding into a system, then $\vec{F}_{thrust}$ will act along $\vec{v}_{rel}$ and value will be λv_{rel}.

→ If mass is leaving a system, then $\vec{F}_{thrust}$ will act opposite to $\vec{v}_{rel}$ and value will be $\left|\frac{dm}{dt}\right| v_{rel}$.

Rocket Propulsion

& No gravity



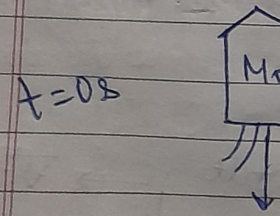
Find velocity of rocket at any time t.

In 1 sec → λ kg

t sec → λt kg

∴ Mass of Rocket after t sec = $M_0 - \lambda t$

As mass is not const, accn might not be constant.



Using NL on rocket at t sec: ($F=ma$)

$$\lambda v = (M_0-\lambda t)\frac{dv}{dt}$$

$$= \lambda v \int_0^t \frac{dt}{M_0-\lambda t} = \int_0^v dv$$

$$= \frac{\lambda v}{-\lambda}\ln(M_0-\lambda t)_0^t = {}^{0}(v)_0^v$$

$$= -u(\ln(M_0-\lambda t) - \ln(M_0)) = v \Rightarrow \boxed{v = u\ln\left(\frac{M_0}{M_0-\lambda t}\right)}$$

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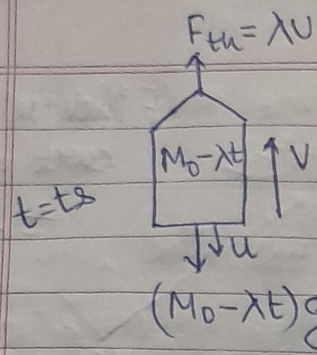
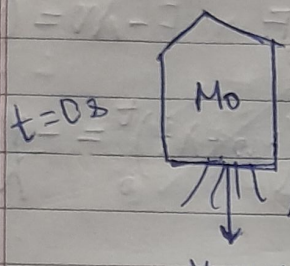


Consider gravity = const
Find velocity of rocket at any time t.

Using NL on rocket at t sec:

$$\lambda v-(M_0-\lambda t)g=(M_0-\lambda t)\frac{dv}{dt}$$

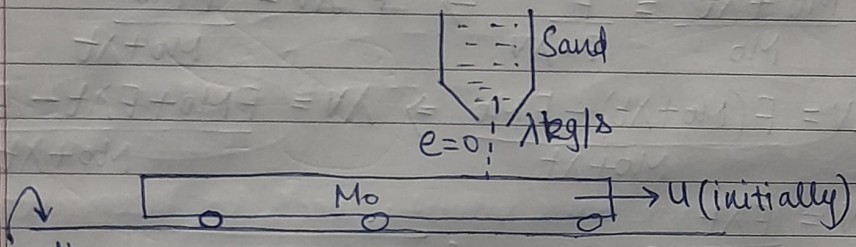
$$=\lambda u\int_0^t\frac{dt}{M_0-\lambda t}-g\int_0^t dt=\int_0^v dv$$



$$=-u\ln\left(M_0-\lambda t\right)_0^t-g(t)_0^t=(v)_0^v$$

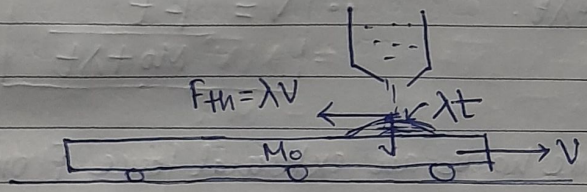
$$\Rightarrow\boxed{v=u\ln\left(\frac{M_0}{M_0-\lambda t}\right)-gt}$$

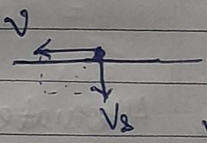
Q



Find velocity of cart at any time t.

At t=ts





As sand falls on cart it has velocity v_s wrt ground

To find v wrt cart we must add v in opp dirⁿ to v_s
Now v_s will be balanced by N of ground

$\therefore$ v is the only rel velocity which changes speed of cart

$F_{th}=\lambda v$

Using NL: $-\lambda v=(M_0+\lambda t)\frac{dv}{dt}$ (-ve as acc^n is $\perp$ vel)

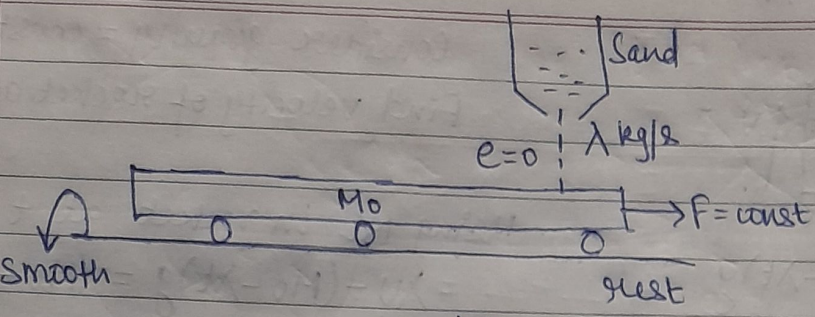
$$=-\lambda\int_0^t\frac{dt}{M_0+\lambda t}=\int_u^v\frac{dv}{v}\Rightarrow-\ln(M_0+\lambda t)_0^t=\ln(v)_u^v$$

$$=\ln\left(\frac{M_0}{M_0+\lambda t}\right)=\ln\left(\frac{v}{u}\right)\Rightarrow\boxed{v=\frac{M_0u}{M_0+\lambda t}}$$

Alternate: M/C along Hor^z of system $\Rightarrow M_0u+\lambda t(0)=(M_0+\lambda t)v$

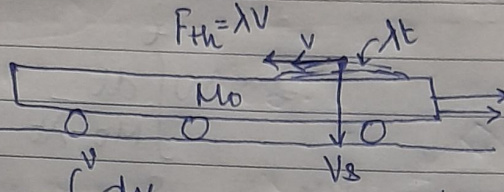
$$\boxed{v=\frac{M_0u}{M_0+\lambda t}}$$

Q



Find velocity of cart any time t.

At t=t sec



Using NL:

$$F-\lambda V=(M_o+\lambda t)\frac{dv}{dt}$$

$$=\int_0^t \frac{dt}{M_o+\lambda t}=\int_0^v \frac{dv}{F-\lambda v} \Rightarrow \frac{1}{\lambda}\ln(M_o+\lambda t)_0^t=-\frac{1}{\lambda}\ln(F-\lambda v)_0^v$$

$$=\ln\left(\frac{M_o+\lambda t}{M_o}\right)=-\ln\left(\frac{F-\lambda V}{F}\right)\Rightarrow \ln\left(\frac{M_o+\lambda t}{M_o}\right)=\ln\left(\frac{F}{F-\lambda V}\right)$$

$$=\frac{M_o+\lambda t}{M_o}=\frac{F}{F-\lambda v} \Rightarrow F-\lambda V=\frac{FM_o}{M_o+\lambda t}$$

$$=\lambda V=\frac{F(M_o+\lambda t)-FM_o}{M_o+\lambda t} \Rightarrow \lambda V=\frac{FM_o+F\lambda t-FM_o}{M_o+\lambda t}$$

$$=\lambda V=\frac{F\lambda t}{M_o+\lambda t}$$

$$\therefore \boxed{V=\frac{Ft}{M_o+\lambda t}}$$

Alternate: As we have an external force F on cart we cannot conserve momentum of system. But we can use IMT.

Using IMT on system $\Rightarrow$ $J=\Delta P=P_f-P_i$

$$Ft=(M_o+\lambda t)v-0$$

$$\therefore \boxed{V=\frac{Ft}{M_o+\lambda t}}$$

Ex